

**BIOINFORMATICS RESEARCH TRENDS IDEATION
USING TOPIC MODELING**



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TO

The Holy Prophet Hazrat

MUHAMMAD

Peace Be Upon Him and His Family

He is greatest Teacher of the World

AND

Parents and Friends

Whose hands always raised in prayers for me and who always with me

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First of all, due praises to Almighty "**ALLAH**", the most merciful and compassionate, the most gracious and beneficent, whose bounteous blessings enable me to perceive and peruse higher ideals of life. All praises and respects are to His Holy Prophet "**MUHAMMAD**" (Peace Be Upon Him and His Family) who enables us to recognize our creator.

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ABSTRACT

Field of bioinformatics is continuously evolving through interdisciplinary research and has resulted in millions of articles available in bioinformatics repositories. Research ideation i.e. process of generating research ideas, is an integral part of researcher's life to proceed in research. Typically, researchers have to access multiple resources to get information about recent research trends. Other sources for research trends identification include scientific literature, online databases and repositories, conferences and workshops, and industry and clinical collaborations. Most of these sources, where the physical participation of the researchers is mandatory, are not feasible for researchers due to time and cost constraints. However, literature database and data repositories subjected to topic modeling are particularly suitable for this purpose due to its ability to uncover the main themes within the large corpus of the data. Presently there are many methods used for topic modeling but Non-Negative Matrix Factorization (NMF) and the Latent Dirichlet Allocation (LDA) are most efficient and popular. As bioinformatics has multiple literature databases (including PubMed, OMIM) which can effectively be used to assess the research trends. So, in this study the published data related to bioinformatics was retrieved from PubMed and was analyzed using LDA and NMF algorithms to uncover the research trends. The output given by this study was sequence analysis, next generation sequence analysis, and virtual screening. The results were interpreted and confirmed by thorough analysis. This study will help biologists with research interests in domain of biology and bioinformatics to identify the current and future research direction of the field.

CHAPTER NO 1

INTRODUCTION

1.1 Background and Motivation

Bioinformatics, an interdisciplinary field that consists of biology, computer science, and information technology, has emerged as a crucial domain in modern scientific research. With the rapid advancement of high-throughput technologies and the exponential growth of biological data, bioinformatics has become an essential tool for analyzing, interpreting, and extracting valuable insights from vast amounts of complex data. The integration of computational methods and biological knowledge has revolutionized our understanding of biological systems, paving the way for groundbreaking discoveries and innovations in areas such as genomics, proteomics, drug discovery, and personalized medicine.

One of the primary challenges faced by researchers in the field of bioinformatics is keeping up with the ever-increasing volume of biomedical literature. As new research findings are continuously published, it becomes increasingly difficult for researchers to stay abreast of the latest developments, emerging trends, and novel topics within their areas of interest. The sheer magnitude of available literature, coupled with the interdisciplinary nature of bioinformatics, compounds this challenge, making it arduous for researchers to efficiently navigate and synthesize the vast knowledge base.

The biomedical literature is a rich repository of scientific knowledge, capturing the collective efforts and discoveries of researchers worldwide. PubMed, maintained by the National Center for Biotechnology Information (NCBI), is a comprehensive database that serves as a central hub for biomedical literature. As of 2024, PubMed encompasses over 37 million citations from biomedical literature, including journals, books, and other sources. This vast and continuously expanding resource presents both opportunities and challenges for researchers seeking to extract meaningful insights and identify emerging research trends.

Manually sifting through and analyzing such a vast corpus of literature is an overwhelming and time-consuming task, often leading to inefficiencies and potential missed opportunities. Researchers may inadvertently overlook crucial findings or fail to recognize emerging patterns and connections within the data, hindering their ability to stay at the forefront of their field. Moreover, the interdisciplinary nature of bioinformatics further compounds this challenge, as researchers must navigate literature spanning various domains, including biology, computer science, statistics, and medicine. There is an urgent need for automated

and effective techniques to get relevant data from the enormous body of biomedical literature in order to tackle these issues. Within huge text corpora, topic modeling, a family of unsupervised machine learning approaches, has proven to be an effective tool for evaluating and revealing hidden patterns and themes. By identifying latent topics and their distributions across documents, topic modeling algorithms offer researchers a means to gain valuable insights, uncover emerging research trends, and explore the relationships between different areas of study.

1.2 Topic Modeling

Topic modeling is the unsupervised machine learning technique of getting the main theme or the idea from large collection of unstructured data. It is very useful when we have many documents and we want to know what topics are discussed without reading each of the document because this can be very time consuming. Topic modeling can give us cluster of main topics discussed in the documents. The use of topic modeling in this research is the same to get the recent research trends in field of bioinformatics in the form of topics. Topic modeling techniques discover underlying semantic structures and group documents into coherent subjects by examining word co-occurrence patterns and distributions across documents. The topic modeling presentation comes next.

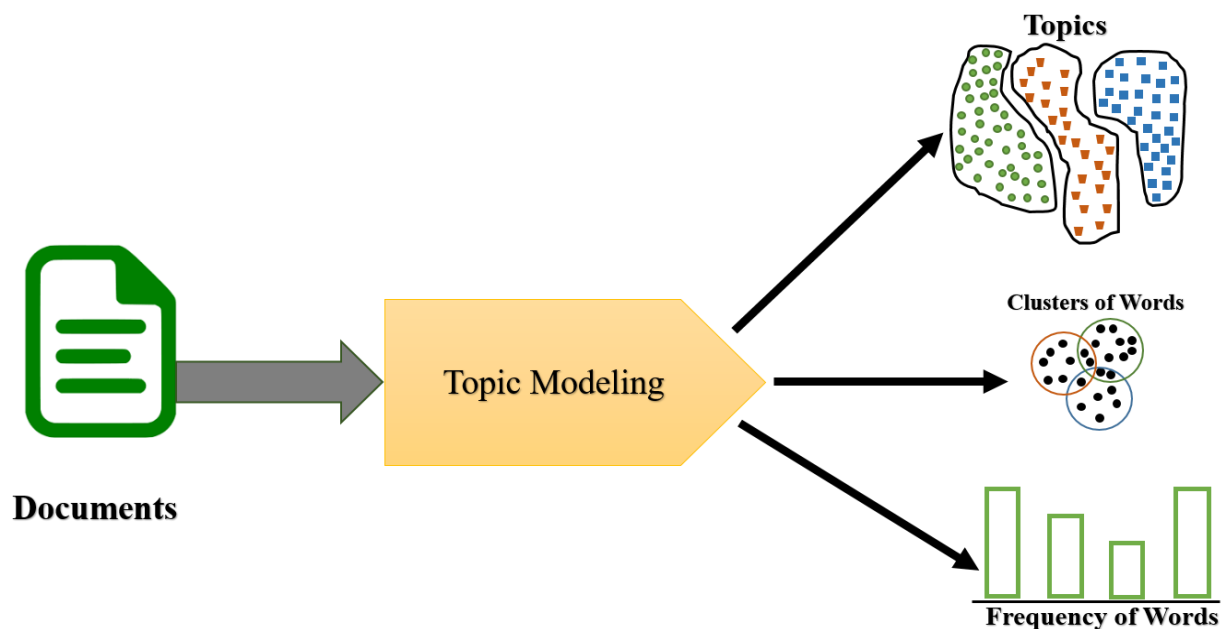


Figure 1.1 Topic Modeling

Documents contain mixes of latent topics, and each subject is represented by a distribution across words. This is the basic idea behind topic modeling. (Kherwa and Bansal, 2019). This approach allows for the extraction of meaningful themes and concepts from text data

without the need for explicit labeling or manual categorization. Instead, the algorithms leverage the inherent statistical patterns within the text to reveal the underlying topics and their relationships.

Topic modeling has found numerous applications across various domains, including computer science, social sciences, digital humanities, and, notably, bioinformatics. In the field of bioinformatics, topic modeling offers a valuable tool for exploring and understanding the vast corpus of biomedical literature, enabling researchers to gain insights into emerging research trends, identify areas of focus, and uncover connections between different fields of study. Non-Negative Matrix Factorization (NMF), Latent Dirichlet Allocation (LDA), Latent Semantic Analysis (LSA), Probabilistic Latent Semantic Analysis (pLSA), Dynamic Topic Models (DTM), Hierarchical Dirichlet Process (HDP), Correlated Topic Model (CTM), BERTopic, and Word Embedding-based Models (e.g., Word2Vec, Doc2Vec) are just a few of the many algorithms used in topic modeling. Selecting an alternative algorithm is contingent upon the particular attributes of the dataset and the goals of the research. (Vayansky and Kumar, 2020).

One of the best advantages of topic modeling is its ability to handle large and diverse text corpora effectively. As the volume of biomedical literature continues to grow exponentially, traditional methods of manually analyzing and synthesizing information become increasingly challenging and time-consuming. Topic modeling algorithms can process and analyze massive amounts of text data, extracting meaningful topics and providing researchers with a high-level overview of the thematic landscape.

Furthermore, topic modeling techniques can reveal hidden relationships and patterns that may not be immediately apparent through manual inspection or keyword-based searches. By identifying co-occurring words and their distributions across documents, these algorithms can uncover semantic connections and highlight interdisciplinary links, fostering a deeper understanding of the interplay between different areas of research. This study intends to use topic modeling technique to systematically analyze large corpus of PubMed research articles, identifying common themes, emerging trends, and areas of ongoing research in the field of bioinformatics. PubMed research articles from last ten years were used in this research and the data included in this research is the topic and abstract of the of the articles (Sperandeo *et al.*, 2020).

However, topic modeling has its own set of difficulties as well. Interpreting and assessing the themes that are generated presents a big problem. Although latent ideas can be found by the algorithms, the concepts that emerge from them might not always be understandable

or consistent with human knowledge. (Layman *et al.*, 2016). This issue can be exacerbated in specialized domains like bioinformatics, where domain-specific knowledge and expertise are often required to accurately interpret and validate the generated topics.

Another challenge lies in the selection and parameterization of appropriate topic modeling algorithms. Different algorithms, such as LDA and NMF, have their own strengths, weaknesses, and underlying assumptions. Choosing the most suitable algorithm and fine-tuning its parameters can significantly impact the quality and interpretability of the resulting topics.

1.3 PubMed and Bioinformatics Literature

One of the best resources for biomedical literature is PubMed, a database kept up to date by the National Center for Biotechnology Information (NCBI) at the National Library of Medicine (NLM).. Serving as a central repository for scientific publications spanning various domains, including bioinformatics, genetics, molecular biology, and clinical sciences, PubMed has become an indispensable tool for researchers worldwide.

Established in 1996, PubMed has experienced remarkable growth, currently indexing over [current number] million citations from biomedical literature, including journals, books, and online resources. This vast and ever-expanding corpus of knowledge encompasses a wide range of disciplines, reflecting the interdisciplinary nature of bioinformatics and its intersection with various fields of study.

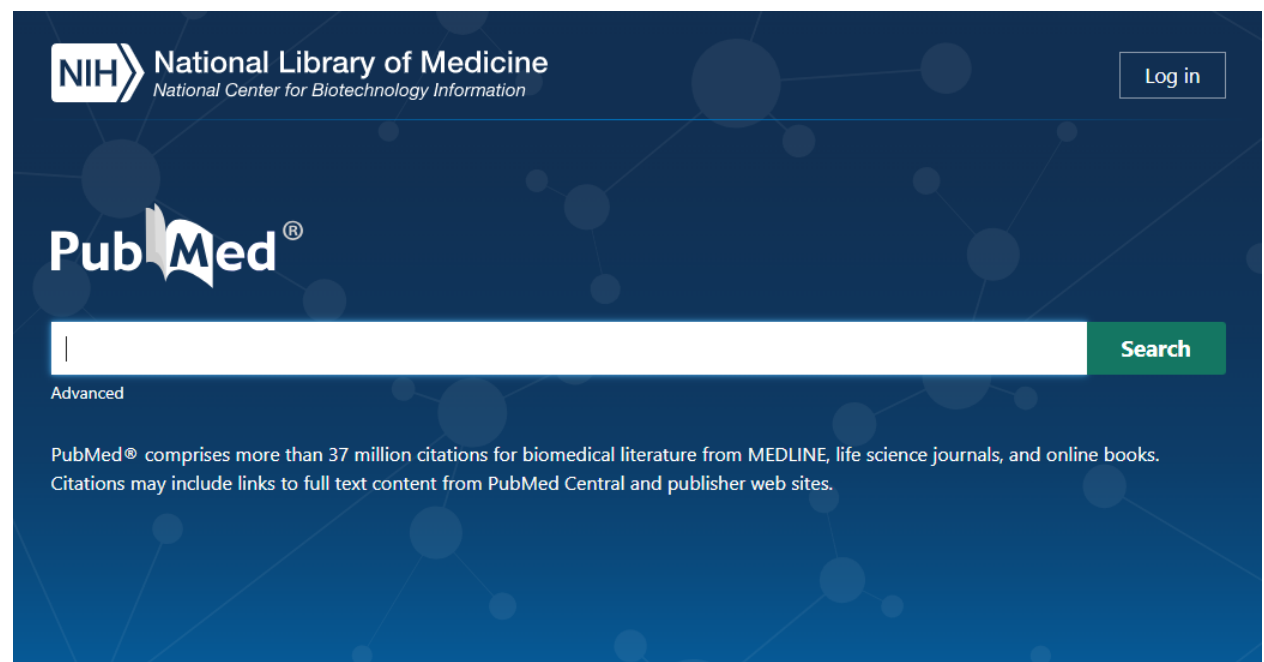


Figure 1.2 PubMed

The significance of PubMed in bioinformatics research cannot be overstated. As a comprehensive database, it provides researchers with access to a wealth of information, facilitating the exploration and synthesis of existing knowledge, identification of research gaps, and the formulation of new hypotheses. By enabling efficient literature searches and retrieval of relevant publications, PubMed empowers researchers to remain current on the most recent advancements, techniques, and discoveries in their fields of study.

However, the sheer volume and complexity of the data contained within PubMed present substantial challenges for researchers seeking to extract meaningful insights and uncover emerging trends. With an ever-increasing number of publications spanning diverse topics and disciplines, manually analyzing and synthesizing the vast corpus of literature becomes an arduous and time-consuming task, often leading to inefficiencies and potential missed opportunities.

The interdisciplinary nature of bioinformatics further compounds this challenge. Researchers must navigate literature spanning various domains, including biology, computer science, statistics, and medicine, each with its own terminology, conventions, and specialized knowledge. Identifying connections and patterns across these diverse fields can be a daunting task, hindering the ability to gain a comprehensive understanding of the research landscape.

Moreover, the rapid pace of scientific discovery and the constant influx of new publications exacerbate the challenges associated with staying abreast of the latest developments. Researchers face the risk of overlooking crucial findings or failing to recognize emerging patterns and trends, potentially impeding their ability to contribute to the forefront of their field.

To solve these issues, there is an urgent need for automated and fast methods to extract useful information from the large corpus of biological literature within PubMed. By leveraging advanced techniques such as topic modeling, researchers can gain valuable insights, uncover emerging research trends, and explore the relationships between different areas of study, ultimately advancing the field of bioinformatics and contributing to scientific progress.

1.4 Topic Modeling Algorithms

In the realm of topic modeling, two prominent algorithms have gained widespread recognition and adoption: Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). These algorithms offer distinct approaches and underlying

assumptions for uncovering latent topics and themes within large text corpora, making them valuable tools for analyzing the vast repository of biomedical literature in PubMed.

1.4.1 Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation (LDA) is a generative probabilistic framework that has become a popular method for topic modeling. Developed by David Blei, Andrew Ng, and Michael Jordan in 2003, LDA functions along the idea that each document in a corpus is a combination of latent themes, and each subject is represented by a distribution of words. The underlying principles of the LDA algorithm is that topics are generated by first sampling a distribution over documents, and then sampling words from each topic based on their respective probabilities (Kumar *et al.*, 2023). This generative approach enables LDA to automatically identify underlying topics in a corpus of texts and assign relevant topic distributions to each document. (Bastani *et al.*, 2019).

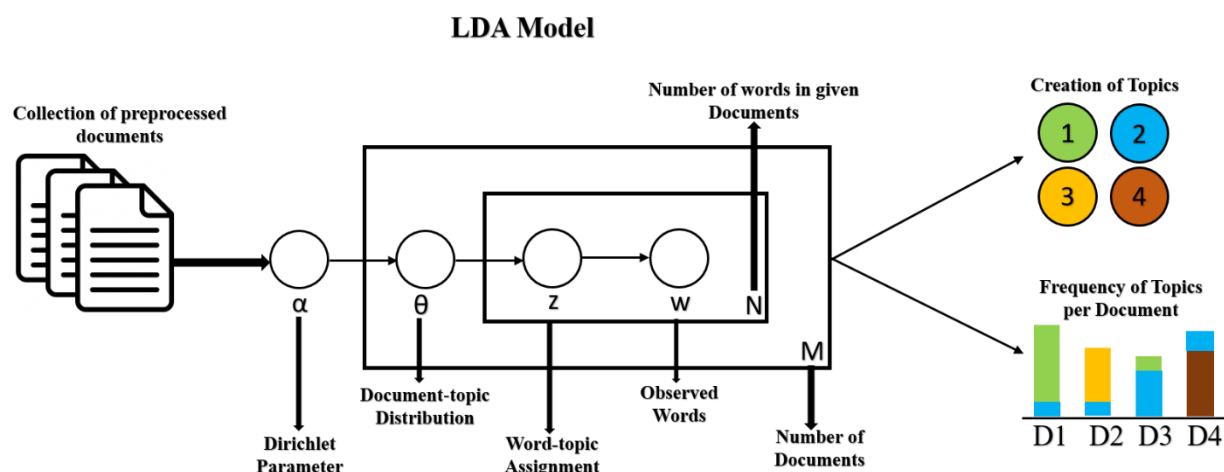


Figure 1.3 Latent Dirichlet Allocation LDA

One of LDA's primary advantages is the capability to deal with the polysemy which occurs when a word has numerous meanings based on its context. By modeling topics as distributions over words, LDA can successfully identify the semantic links between words and subjects, enabling a more detailed comprehension of the underlying themes. Additionally, LDA's probabilistic framework allows for the incorporation of prior knowledge and enables further extensions and modifications to the model.

However, LDA also has some limitations. It assumes that the number of topics is known a priori, which can be challenging to determine in practice, especially for large and diverse text corpora like biomedical literature. Additionally, LDA can struggle with identifying and separating highly correlated topics, as it assumes that topics are independent of each other (Williams *et al.*, 2020).

1.4.2 Non-negative Matrix Factorization (NMF)

NMF is another effective topic modeling approach, especially for obtaining additive parts-based representations for non-negative data. Unlike LDA, which is a generative probabilistic model, NMF is a matrix factorization method that divides a non-negative matrix (representing document-term occurrences) into two separate non-negative matrices: one for the topic-term distributions and one for the document-topic distributions. NMF operates under the assumption that each document is a linear combination of topics, and each topic is represented by a non-negative vector of word weights. This additive nature of NMF makes it particularly useful for capturing the inherent structure of the data, as it ensures that the resulting topics and their representations are interpretable and easily understandable.

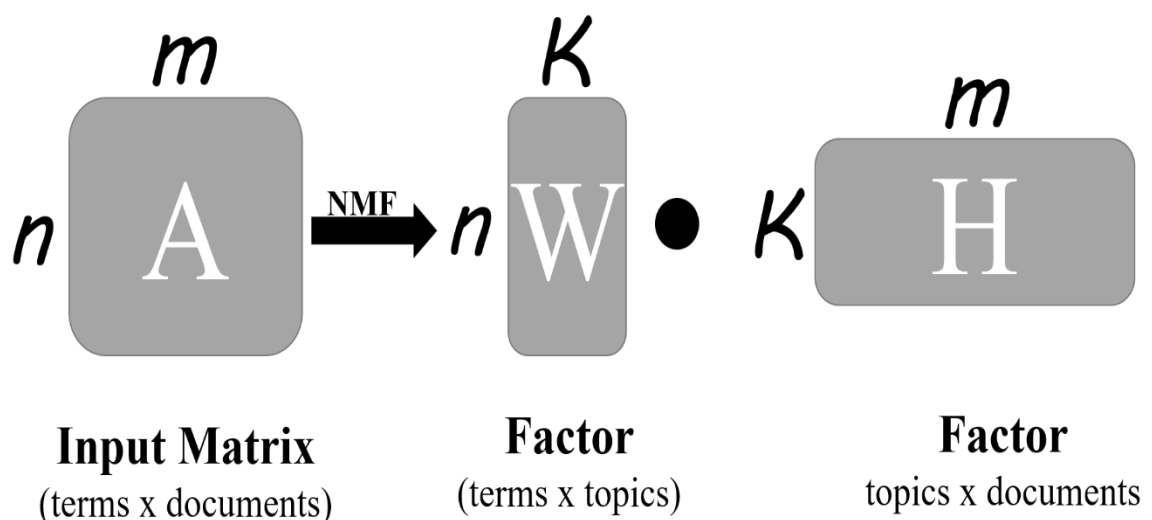


Figure 1.4 Non-negative Matrix Factorization (NMF)

One of the key advantages of NMF is its ability to handle sparse data effectively, which is a common characteristic of text corpora. Additionally, NMF does not require the specification of the number of topics a priori, as it can be determined through various techniques, such as exploring the reconstruction error or employing sparsity constraints. However, NMF also has its limitations. Unlike LDA, which provides a probabilistic interpretation of topics and their distributions, NMF lacks this explicit probabilistic framework, making it harder to incorporate prior knowledge or to extend the model in certain ways. Furthermore, NMF can struggle with capturing the underlying semantics of the data, as it primarily focuses on the co-occurrence patterns of words within documents.

1.5 Programming Language, Tools and Libraries

In the pursuit of uncovering latent topics and emerging research trends within the vast corpus of biomedical literature from PubMed, this study leverages a powerful combination of tools and libraries. At the core of the computational workflow lies Jupyter Notebook, a versatile and widely adopted platform for interactive computing and data analysis.

1.5.1 Python and its Importance in Topic Modeling Research

Python is essential for topic modeling research because of its extensive libraries, easy to use, and robust support of the community. Key libraries, such as Gensim and scikit-learn, give robust implementations of the algorithms like LDA, making model building and training easier. Python's integration with tools like Jupyter Notebooks and its powerful data manipulation libraries (Pandas, NumPy) make data preprocessing and analysis easier. Visualization libraries such as Matplotlib and PyLDAvis can also help with effective result interpretation and presentation. Python is the language of choice for researchers in topic modeling due to its flexibility and extensive resources.

1.5.2 Jupyter Notebook: A Powerful Interactive Computing Environment

Jupyter Notebook is a web-based open-source program that enables academics to generate and share notebooks that include live code, graphics, and narrative text. This collaborative computing platform seamlessly integrates code, results, and explanations, promoting collaboration, reproducibility, and quick prototyping.

Jupyter Notebook supports a variety of computer various languages, like Python, R, and Julia, amongst more. Python was chosen as the major programming language for this study because of its large ecosystem of packages for natural language processing (NLP), machine learning (ML), and data analysis.

Within the Jupyter Notebook environment, researchers can seamlessly combine code execution, data exploration, and result visualization, facilitating an iterative and exploratory approach to data analysis. This interactive nature is particularly beneficial in the context of topic modeling, where researchers can do experiments with the different algorithms, parameters, and preprocessing methods, and immediately observe the impact on the generated topics and their interpretability.

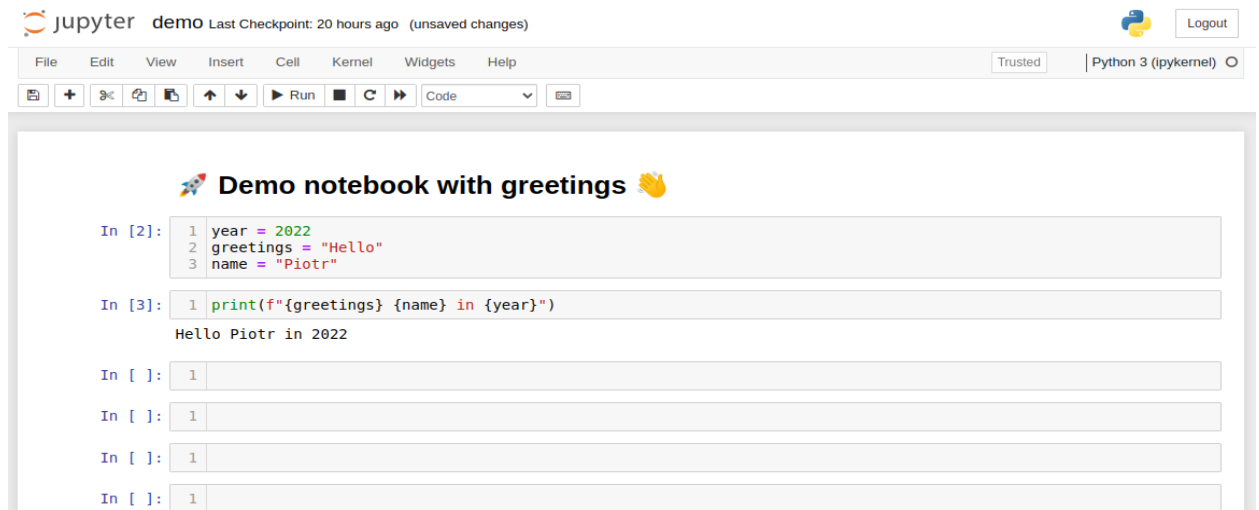


Figure 1.5 Jupyter Notebook

1.5.3 Python Libraries for Natural Language Processing and Topic Modeling

To unlock the potential of topic modeling and uncover the hidden themes within the biomedical literature, this study leverages a range of powerful Python libraries, each designed to tackle specific aspects of the data analysis pipeline.

1.5.3.1 Natural Language Toolkit (NLTK):

The Natural Language Toolkit (NLTK) is a comprehensive collection of Python tools and programs for natural language processing. In this work, NLTK is used for text preparation tasks such as tokenization, and stop word removal, stemming, and lemmatization. These preparation processes are essential for putting the unprocessed data into a format appropriate for topic modeling algorithms. (Saxton, 2018) cd. Key NLTK modules utilized in this study include:

- [nltk.corpus](#): Provides access to corpora, including stop word lists.
- [nltk.tokenize](#): Offers tokenization functionality for breaking text into words or other meaningful units.
- [nltk.stem](#): Contains stemming and lemmatization algorithms for reducing words to their base or root forms.

1.5.3.2 Gensim:

Gensim is a strong and scalable open-source library for unsupervised topic modeling, document indexing, and similarity retrieval in Python (Saxton, 2018). This study leverages Gensim's implementations of popular topic modeling algorithms, such as LDA and NMF. Key Gensim modules utilized in this study include:

- [gensim.models.LdaMulticore](#): Provides an efficient implementation of the LDA algorithm, supporting parallel computation for improved performance.
- [gensim.models.NmfModel](#): Offers an implementation of the Non-negative Matrix Factorization algorithm for topic modeling.
- [gensim.corpora](#): Provides functionality for creating and manipulating document corpora, essential for topic modeling tasks.
- [gensim.models.CoherenceModel](#): Enables the evaluation of topic model coherence, a measure of the semantic interpretability of the generated topics.

1.5.3.3 Scikit-learn:

Scikit-learn is a Python machine learning framework that includes a diverse set of techniques and instruments for data extraction and analytics. In the context of this study, scikit-learn is utilized for various tasks, including text preprocessing, feature extraction, and dimensionality reduction (Kumar and Gupta, 2024). Key scikit-learn modules employed in this study include:

- [sklearn.feature_extraction.text](#): Offers utilities for converting text data into numerical feature vectors suitable for machine learning algorithms.
- [sklearn.decomposition](#): Provides implementations of matrix decomposition algorithms, such as LDA and NMF, which is used for topic modeling.

1.5.3.4 pyLDavis:

pyLDavis is a Python library designed to create interactive conceptions for the topic models, particularly those generated using the LDA algorithm. This study employs pyLDavis to visualize and interpret the generated topics, enabling researchers to gain insights into the thematic structure of the biomedical literature corpus. Key functionalities of pyLDavis utilized in this study include:

- Visualizing the inter-topic distance map, which represents the similarities and relationships between topics.
- Generating term frequency visualizations for individual topics, allowing researchers to examine the most salient terms within each topic.
- Creating interactive term frequency visualizations across topics, enabling the exploration of term distributions and their associations with multiple topics.

These visualizations provided by pyLDavis play a crucial role in interpreting and validating the generated topics, facilitating a deeper understanding of the underlying

themes and their connections within the biomedical literature corpus (Onah and Pang, 2021).

1.5.3.5 Additional Libraries:

Apart from the aforementioned libraries, this study also incorporates several other Python packages to enhance the data analysis and visualization capabilities:

- [wordcloud](#): A library for generating word clouds from text data, providing a visually engaging representation of the most prominent terms within the corpus or topics.
- [seaborn](#) and [matplotlib](#): Popular data visualization libraries for creating informative and visually appealing plots, charts, and graphs, enabling the effective communication of research findings.
- [tqdm](#): A progress bar library that provides visual feedback on the progress of long-running operations, enhancing the user experience during computationally intensive tasks.
- [re](#): The regular expressions library, used for advanced text processing and pattern matching tasks within the biomedical literature corpus.

By leveraging this powerful combination of tools and libraries within the Jupyter Notebook environment, this study aims to streamline the data analysis workflow, facilitate reproducibility, and ultimately uncover meaningful insights into the recent bioinformatics research trends by effectively modeling and interpreting the topics present in the vast corpus of biomedical literature from PubMed.

1.6 Research Objectives and Contributions

The overarching aim of this study is to leverage the power of the topic modeling techniques to uncover emerging research trends and themes within the rapidly expanding field of the bioinformatics. By analyzing the vast corpus of biomedical literature available in PubMed, this research aims to provide researchers and medical practitioner with a comprehensive thoughtful of the current landscape and future directions in this interdisciplinary domain.

Objectives

- To apply topic modeling algorithms e.g. LDA, NMF on bioinformatics research articles.
- To recognize and categorize current trends and topics in bioinformatics research.
- To assess the effectiveness of LDA and NMF in capturing and summarizing various themes found in PubMed data.

Through the achievement of these objectives, this study aims to make significant contributions to the field of bioinformatics by providing a comprehensive and data-driven understanding of recent research trends. By leveraging advanced topic modeling techniques and harnessing the wealth of information contained within PubMed, this research has the potential to uncover valuable insights, foster interdisciplinary collaborations, and drive innovation in this rapidly evolving domain.

1.7 Significance of the Proposed Study

The proposed study holds significant importance and potential impact in the field of bioinformatics and beyond. By leveraging advanced topic modeling techniques to analyze the vast corpus of biomedical literature available in PubMed, this research aims to uncover valuable insights and facilitate the efficient extraction of knowledge from this rapidly growing and diverse domain.

1.7.1 Unveiling Emerging Research Trends and Topics:

The ability to identify and characterize emerging research trends and topics in bioinformatics is crucial for researchers, practitioners, and policymakers alike. This study's approach to topic modeling will provide a data-driven perspective on the current landscape and future directions of the field, enabling stakeholders to make informed decisions and allocate resources effectively. By uncovering the prominent themes and areas of focus, this research can guide the development of new research initiatives, foster interdisciplinary collaborations, and drive innovation in this rapidly evolving domain.

1.7.2 Facilitating Knowledge Discovery and Dissemination:

The biomedical literature represents a rich repository of knowledge, but navigating and synthesizing this vast corpus can be a daunting task. Topic modeling techniques can serve as powerful tools for knowledge discovery, enabling researchers to efficiently explore and understand the relationships between different subfields, identify potential cross-pollination opportunities, and gain a comprehensive understanding of the state-of-the-art in their areas of interest. By providing interpretable representations and visualizations of the identified topics, this study will facilitate the effective communication and dissemination of research findings, fostering knowledge sharing and collaboration within the bioinformatics community.

1.7.2 Interdisciplinary Insights and Collaborations:

Bioinformatics is an inherently interdisciplinary field, drawing upon knowledge and methodologies from various domains, including biology, computer science, statistics, and

medicine. This study's focus on examining the connections and overlaps between topics spanning different disciplines will shed light on the convergence of ideas and the potential for cross-fertilization of methods and techniques. By revealing these interdisciplinary links, the research can stimulate collaborations between researchers from diverse backgrounds, fostering innovative approaches and breakthroughs in addressing complex biological problems.

1.7.3 Anticipating Future Directions and Aligning Research Efforts:

Accepting the sequential changing aspects and evolution of research topics is essential for anticipating future trends and aligning research efforts accordingly. This study's analysis of the delivery of themes over time will enable researchers and stakeholders to identify emerging areas of interest, as well as potential areas that may be declining in relevance. This knowledge can inform strategic decision-making processes, such as resource allocation, funding priorities, and the improvement of learning courses and curricula tailored to evolving needs of the field.

1.7.4 Enhancing Computational Methodologies and Tools:

The proposed study will contribute to the advancement of computational methodologies and tools for topic modeling and natural language processing within the context of biomedical literature analysis. By developing customized preprocessing techniques, incorporating domain-specific knowledge, and fine-tuning algorithms for this specialized domain, the research will pave the way for more robust and effective topic modeling frameworks tailored to the unique challenges and complexities of bioinformatics data.

By addressing these significant aspects, the proposed study has the potential to make substantial contributions to the field of bioinformatics, fostering scientific progress, facilitating knowledge sharing, and driving innovation in this rapidly evolving and interdisciplinary domain.

CHAPTER NO 2

REVIEW OF LITERATURE

Liu *et al.* (2006) investigated the application of nonnegative matrix factorization (NMF) in pattern recognition. The study looked at several applications where NMF proved to be effective, including facial recognition, document clustering, and gene expression analysis. Through an extensive review of existing literature and experimental demonstrations, the researchers highlighted the benefits and drawbacks of using NMF techniques in pattern recognition tasks. They explained how NMF allows for the extraction of meaningful features from high-dimensional data while enforcing non-negativity constraints, improving interpretability and performance in pattern recognition tasks. The paper also discussed algorithmic implementations and offered perspectives on future research directions in this field.

Lin and Wilbur (2007) developed and presented a probabilistic model which gave the similarity found in the content which was called PubMed related articles (PMRA). It was basically an algorithm for content similarity deployed in the PubMed search engine which authors compared with Best Matching 25 (bm25) and resulting in giving significant improvement over bm25. Their methodology included constructing the model which extracts topics present in the included documents, parameter estimation, and results. Through this model more related articles could be found in the database of PubMed. This model became more effective than the previous ones because it was quick and it helped researchers to enhance the accuracy and efficiency in identifying related articles, thereby facilitating more effective information retrieval and knowledge exploration within the biomedical literature.

Zheng *et al.* (2014) proposed an autoregressive approach to multimodal topic modeling. They sought to address the challenge of effectively analyzing data in multiple modalities, including text, images, and audio. Using autoregressive techniques, the authors created a novel topic modeling methodology that captures the interdependence of different modalities. Their approach is a promising solution for detecting latent topics in multimodal datasets, allowing for deeper insights and more comprehensive analyses. Zheng *et al.* helped advance the field of computer vision and pattern recognition by presenting their findings at a prestigious conference, providing a valuable framework for researchers looking to investigate complex multimodal data sets.

Anupriya and Karpagavalli (2015) endeavored to analyze the journal abstracts by implementing LDA with Collapsed Variational Bayesian Optimization (CVBO) and Gibbs sampling techniques to explore the underlying topics. A dataset with 200 abstracts within four topics of machine learning, cardiovascular disease in the field of medical, computer network in the field of computer science and cancer from two different domains of computer science and medical journal was used. Their research methodology included defining LDA method with CVBO Sampling and Gibbs Sampling, dataset, preprocessing of the data (Scala NLP API used), and document modeling. Their main aim was to recommend appropriate tags for journal abstracts.

Pyo *et al.* (2015) investigated a novel strategy for comparable TV (television) user categorization and TV program system recommendations, known as Social TV. The writers used TV poll agency data gathered over a seven-month period, including TV reviews of previous data and electronic system guide data containing program term definitions. They employed two LDA models: among in the topic-specific modeling of TV users and another for topic modeling of description terms for watched TV shows. The authors then combined the two LDA topic models with various parameters to produce similar TV consumers and related TV program summary terms at the same time. Their unified model then determined the semantic relationship between TV user communities and TV program summary word categories, allowing for more precise TV program suggestions. By doing so they overcame a problem of recommending new programs to TV users. Authors also compared their proposed model (TMud) with BLTM and PLTM which found out to be superior than BLTM and PLTM.

Vijayarani *et al.* (2015) published in the International Journal of Computer Science and Communication Networks. Authors provided a thorough overview of preprocessing techniques for text mining. They investigate various methods for improving the effectiveness of text mining processes. The researchers look at techniques like tokenization, stemming, stop-word removal, and normalization, among others. Through a critical examination of each technique, they highlight their respective roles in improving the quality and efficiency of text mining results. The paper also discusses how preprocessing improves the accuracy of text mining algorithms and models. By synthesizing existing literature and methodologies, the authors shed light on the preprocessing phase of text mining, contributing to a better understanding of this critical aspect of natural language processing.

Layman *et al.* (2016) investigated the use of topic modeling for analyzing NASA space system problem reports. They used Latent Dirichlet Allocation (LDA) to classify and

comprehend the themes in these reports. The study aimed to identify frequent problems and patterns in problem reports in order to improve software maintenance and development processes. The authors evaluated LDA's effectiveness in extracting meaningful topics from data, taking into account both practical applications and methodological considerations. Their findings highlighted topic modeling's potential for improving software repository management. The study also shed light on the challenges and limitations of using LDA in this context. The study showed that combining advanced data mining techniques with domain-specific knowledge can help software engineers make better decisions and solve problems.

Liu *et al.* (2016) presented a review in which they gave us the basic difference between the traditional clustering and using topic models in biological data. With the background knowledge of topic model, discussion of topic modeling was also included in which steps related to topic modeling like the BoW (Bag of Words), model training, and model output were illustrated. The development of topic model, extension model for LDA were discussed, and toolkits used for topic models like genism, Stanford topic modeling toolbox (TMT), MALLET, and some other open source software were briefed. This study described the usage of topic modeling in bioinformatics including different articles selectin, the topic modeling tasks for bioinformatics, significance for biological-data clustering and classification.

Syed and Spruit (2017) explored the efficacy of full-text versus abstract data for estimating topic coherence scores employing Latent Dirichlet Allocation (LDA). They investigated the relationship between the completeness of data representation and the accuracy of topic modeling results. The researchers sought to identify any significant differences in LDA model performance by analyzing coherence scores from both full-text and abstract datasets. Their research aimed to shed light on the best data granularity for topic modeling tasks, providing insights that could be used to develop more efficient and effective text analysis methodologies. Syed and Spruit contributed to the ongoing discussion about the methodological considerations inherent in topic modeling research by conducting rigorous experiments and analyses, enriching our understanding of LDA's application in data science and advanced analytics.

Gerlach *et al.* (2018) explored topic modeling with network analysis. The problems with the traditional topic modeling and their possible solutions have been discussed. This research investigated the implementation of network analysis techniques with the traditional topic modeling methods to enhance the understanding of complex relationships

within textual data. The authors accomplished this by modeling text corpora as bipartite networks of documents and words. By implementing existing community-detection methods, they were able to create a more versatile topic modeling framework that automatically detected the number of topics and clustered both words and documents. They also compared LDA and hSBM on both real and artificial data. Their research focused on the application of hSBM to Wikipedia articles. Materials and method were also discussed which included the minimum description length for both models, artificial corpora, datasets for real corpora, and numerical implementations.

Saxton (2018) provided an understandable overview of topic modeling techniques designed specifically for theological librarianship. Saxton explained the fundamental principles of topic modeling, emphasizing its usefulness in managing and analyzing large amounts of textual data. The paper described how to implement Latent Dirichlet Allocation (LDA) in Python and provided practical examples to demonstrate the process. Saxton's work sought to demystify the technical aspects of topic modeling, making it more accessible to those without extensive computational backgrounds. As a result, the paper served as an invaluable resource for librarians and scholars interested in incorporating advanced text analysis tools into their research and library management practices.

Asmussen and Møller (2019) aimed to facilitate exploratory literature reviews by using topic modeling techniques to efficiently extract and organize key themes from large corpora of scholarly articles. The authors demonstrated the efficacy of their method in identifying latent patterns and trends across various fields of study by employing Latent Dirichlet Allocation (LDA) and other topic modeling algorithms. Using machine learning and natural language processing, the smart literature review framework provided a systematic and data-driven approach to navigating large amounts of literature. The study provided practical guidelines for implementing this methodology, allowing researchers to conduct more comprehensive and insightful literature reviews with increased efficiency and accuracy. Authors innovative approach helped advance literature review methodologies, providing a valuable tool for researchers across domains.

Bastani *et al.* (2019) conducted a study using LDA to model topics within consumer objections give in to to the Consumer Financial Protection Bureau (CFPB). They aimed to identify common themes and trends in these complaints in order to better understand consumer concerns. The researchers examined a large dataset of CFPB complaints, using LDA to identify underlying themes. Their findings shed light on common issues that financial consumers face. The study demonstrated the effectiveness of LDA in processing

and categorizing large amounts of textual data, highlighting its potential to improve consumer protection efforts. Overall, the study made significant methodological contributions to the use of topic modeling in consumer complaint analysis.

Kherwa and Bansal (2019) conducted a comprehensive review of topic modeling, presenting a detailed analysis of various algorithms and their applications in various fields. They concentrated primarily on LDA and its application in mining latent themes from large datasets. The authors investigated the evolution of topic modeling techniques, comparing performance and scalability. They highlighted each method's strengths and limitations, providing insights into its suitability for various types of data. The review also addressed issues regarding topic coherence and interpretability. They also talked about new evaluation metrics for topic models. The paper concluded by outlining potential future research directions for improving the effectiveness of topic modeling in information systems.

Albalawi *et al.* (2020) studied a variety of topic-specific modeling techniques for short written data, including five. The techniques employed included NMF, latent semantic analysis (LSA), LDA, random projection (RP), and principal component analysis (PCA). Two sets of data were utilized to examine the effectiveness of the topic selected for modeling strategies. A few conventional statistical assessment metrics such as recall, F-score, and topic coherence were also used. In paper they started with the introduction along with literature review and then defining the application of topic modeling and gave the toolkits for topic models. After that researchers gave their methodology for the research in which they used data processing and topic modeling methods. In the end result was given which concluded that LDA and NMF gave best results overall than the other methods used in topic modeling of the short texts.

Blair *et al.* (2020) conducted a comprehensive review of topic modeling, presenting a detailed analysis of various algorithms and their applications in various fields. They concentrated primarily on Latent Dirichlet Allocation (LDA) and its application in extracting latent topics from large datasets. The authors investigated the evolution of topic modeling techniques, comparing performance and scalability. They highlighted each method's strengths and limitations, providing insights into its suitability for various types of data. The review also addressed issues regarding topic coherence and interpretability. They also talked about new evaluation metrics for topic models. The paper concluded by outlining potential future research directions for improving the effectiveness of topic modeling in information systems.

Jelodar *et al.* (2020) employed a deep sentiment classification and topic discovery approach to analyze online discussions related to the novel coronavirus, or COVID-19. Utilizing LSTM recurrent neural network models, they conducted natural language processing (NLP) to extract insights from the vast amount of online discourse surrounding the pandemic. The research, published in the IEEE Journal of Biomedical and Health Informatics, focused on understanding the sentiments expressed in COVID-19 discussions and identifying prevalent topics within these conversations. By applying advanced NLP techniques, the authors aimed to provide a deeper understanding of public perceptions and concerns surrounding the pandemic. Their findings contribute to the field of biomedical and health informatics by offering valuable insights into the dynamics of online discourse during global health crises like COVID-19.

Rains *et al.* (2020) carried out examination of public and advancement of communication in research. On behalf of this research they used abstracts from 20,000 National Communication Association (NCA), International Communication Association(ICA), and regional journals articles from a time span of 1918 to 2015. The investigation probed into the thematic contours and transformations within the field of communication, shedding light on its trajectory over nearly a century. For topic modeling of the corpus LDA was used which identified the topics present in the corpus. After labeling the extracted topics they validated them manually to verify the presence in the used article abstracts. Their methodology included collection of data and data analysis which include pre-processing of data, selection of appropriate model, labeling the topic and authentication of topics revealed, and evaluating difference in topics. Finally, the results were presented, which included three research questions: topics for research on communication, topic frequency over a period of time, and topic prevalence by publications type. Overall, this study revealed how communication has been researched over time and across various types of publications.

Sethasathien and Prasertsom (2020) focused on using topic modeling techniques to analyze research project data. The authors wanted to show how Latent Dirichlet Allocation (LDA) could be used to reveal hidden thematic structures in large datasets. They conducted a thorough review of existing methodologies, emphasizing the advantages of incorporating topic modeling into data analytics workflows. The authors presented a case study in which they applied LDA to a dataset containing numerous research project descriptions and successfully identified common themes and trends. Their findings demonstrated the power of topic modeling to improve understanding and organization of research data, thereby

improving the ability to identify key areas of focus and emerging topics. They discussed how their findings could improve research management and decision-making processes. The study emphasized the importance of methodological rigor and offered recommendations for implementing LDA in similar settings. By addressing issues such as data preprocessing and parameter selection, the authors added to the growing body of literature promoting advanced analytical techniques in research data analysis. Their findings demonstrated the transformative power of data analytics on the efficiency and effectiveness of research project evaluations.

Shahbazi and Byun (2020) tackled the challenge of topic modeling in short-text data, where traditional techniques often struggle due to the lack of sufficient word co-occurrence information. They proposed a novel approach that combines Non-negative Matrix Factorization (NMF) with deep reinforcement learning to improve the quality and interpretability of topics discovered in short-text corpora. The authors first provided a comprehensive background on the limitations of existing topic modeling techniques, such as LDA and NMF, when applied to short texts. They then introduced their deep reinforcement learning framework, which optimizes the NMF objective function by learning a more effective mapping between the input data and the latent topic representations. The authors employed a reinforcement learning agent that iteratively explores different topic assignments and receives rewards based on the coherence and interpretability of the resulting topics. This approach allowed the model to learn an optimized topic-word distribution that better captures the semantic relationships and co-occurrence patterns in short texts. The authors evaluated their proposed method on various short-text datasets, including news articles, social media posts, and product reviews. They compared the performance of their approach with traditional NMF and other state-of-the-art topic modeling techniques using metrics such as topic coherence, topic diversity, and topic quality. The results demonstrated that their deep reinforcement learning-based NMF approach outperformed existing methods, achieving higher topic coherence and better capturing meaningful topics in short-text data.

Sperandeo *et al.* (2020) examined the significance of personality within the realm of the mental health by applying topic modeling method on the abstracts published within the database of PubMed in last five years. This research was carried out with the aim of exploring the current relationship of personality and mental well-being. Firstly, data was gathered from PubMed using query on “mental health” and “personality.” for five years starting from 31/12/2012. Secondly, they used LDA method for the topic modeling of

abstracts. After that topics were visualized in the form of hierarchical clustering in which 30 topics were extracted from LDA. Analysis of the emerged clusters was done to check the relationship. At the end they gave conclusion about the topics retrieved from LDA which showed that which topic was studied for how many times. This paper's limitations included the exclusion of exploratory factor analysis, global network analysis, and country-specific paper counts.

Svensson and Blad (2020) investigated the performance of NMF and LDA for theme modeling of Swedish news articles. They recognized the challenges posed by morphologically complex languages like Swedish, where word forms can vary significantly, potentially affecting the quality of the generated topics. The authors first preprocessed a corpus of Swedish news articles, applying methods such as tokenization, stopword removal, and lemmatization to normalize the text data. They then applied both NMF and LDA topic modeling algorithms to the preprocessed corpus, carefully tuning the parameters and leveraging appropriate evaluation metrics to assess the quality and interpretability of the generated topics. The authors compared the performance of NMF and LDA using metrics such as topic coherence, topic diversity, and topic quality, taking into account the unique characteristics of the Swedish language. They found that both NMF and LDA were effective in identifying meaningful topics from the Swedish news corpus, with NMF exhibiting better performance in terms of topic coherence and interpretability. The authors provided detailed analyses and visualizations to illustrate the strengths and weaknesses of each approach, highlighting the impact of morphological complexity on topic modeling performance. They also discussed the implications of their findings for researchers and practitioners working with text data in languages with complex morphological structures, offering insights and recommendations for selecting the most suitable topic modeling technique.

Tighe *et al.* (2020) used 200,000 abstracts from PubMed under the subject "pain" to conduct basic natural language processing (NLP) analyses, which included a vector-embedded model of term definitions available through the "word2vec" method, the LDA, a network analysis, and a model based on deep learning. Their research identified relevant terms like pain and patient along with less frequent but impactful terms like "tmd", "qol", and "andometriosis". Using the word2vec technique, their research discovered terms that were related to "pain" such as "discomfort", "symptom", and "pain-related". The terms "acute" and "chronic" were the majority akin to "nonspecific", "subacute", "persistent", "longstanding", and "long-standing". LDA topic modeling yielded the peak cohesiveness

(0.49) among 40 topics. The network analysis identified three topics, two of which were connected to women's wellness and one to their age. The deep learning model effectively generated multiple distinct abstracts. This allowed them to identify trending topics relevant to their research.

Vayansky and Kumar (2020) reviewed some basic topic modeling methods in which they described the advantages and limitations of those topic modeling methods. Firstly, they gave the basic introduction to topic modeling and then starting from LDA the basic and mostly used approach described the standard topic model approaches which includes Topic models with advanced topic relationships, topic models based on time, and topic models optimized for short text. They stated that although LDA is the flexible and adaptive but it's not always best algorithm for some data having complex relationships so, they presented some other approaches for the topic modeling of the data having complex data relationships. With that they reviewed some topic models which can handle short text as social media text are mostly of short text which makes it unable to topic modeling on them properly. They also looked at some algorithms used for optimizing and determining the parameters in topic modeling, as these algorithms help to produce meaningful results despite the method used for topic modeling. Finally, they produced a decision tree based on the features and outcomes of each model, which can assist someone in determining the best topic modeling approaches for their desired datasets.

Williams *et al.* (2020) explored enhancements to Latent Dirichlet Allocation (LDA) topic modeling by integrating gamma and Simmelian filtration techniques. Their research, presented at the 2020 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM), aimed to address common challenges in LDA, such as topic coherence and relevance. They applied gamma filtration to refine the topic distribution and Simmelian filtration to enhance the detection of meaningful topics by leveraging network structures. The study demonstrated that these combined methods improved the quality and interpretability of the topics generated by LDA models. The authors conducted extensive experiments, showing significant advancements in the coherence and relevance metrics of the topics. Their findings suggested that the incorporation of these filtration techniques could lead to more accurate and insightful results in topic modeling applications. This research contributed to the methodological advancements in LDA, providing a framework for more effective data analysis in social network research.

Amara *et al.* (2021) played a pivot role in assessing the significance of the social media data in disaster tracking, monitoring, and risk management. The main focus of this research was doing topic modeling on Facebook public posts related to COVID-19 because Twitter was mostly used for this type of work but Facebook wasn't used as much as twitter so, They decided to use Facebook in seven languages, which are English (EN), Arabic (AR), Spanish (ES), Italian (IT), German (DE), French (FR), and Japanese (JP). Researchers collected data, preprocessed it, applied LDA, and presented the results in a graph structure. Data was collected from January 1st, 2020 to May 15, 2020, divided into three time frames. The first time frame ran from January to February, the second from March to April, and the third from May 15th to June 15th. Finally, they summed up the results for all three periods. The findings of this study began to highlight the gravity of the impending corona disaster for numerous nations.

Gan *et al.* (2021) conducted a comprehensive examination of the Non-negative Matrix Factorization (NMF) technique. This survey delved into various aspects of NMF, including its theoretical foundations, algorithmic developments, and diverse applications across different fields. The authors provided insights into the evolution of NMF, discussing its strengths and limitations, as well as the challenges encountered in its implementation. Through a systematic analysis of existing literature, the paper offered a valuable overview of NMF's significance in signal processing, image analysis, text mining, and other domains. Overall, the survey served as a treasured supply for investigators and practitioners concerned in understanding principles, and applications of Non-negative Matrix Factorization.

Maier *et al.* (2021) described four major challenges encountered by researchers while applying LDA model to textual data. The four challenges were suitable text collection pre-processing, adequate model parameter selection, including a sufficient number of topics to be produced, model reliability evaluation, and valid topic interpretation. They also gave the proper solution to these challenges. It includes proper implication of LDA by first introducing LDA, describing its background, giving LDA advantages, limitations, and challenges, solutions to the challenges like data dusting and pre-processing of formless data available in text, selecting the quantity of topics and earlier constraints or it can be said parameters, and topic interpretation and validity. Finally, they proposed a methodological approach for topic modeling with LDA, explaining the challenges and solutions, and the data came from an ongoing study where they examine civil-society actors' online interactions about the security of food.

Onah and Pang (2021) conducted a study on the design principles of Massive Open Online Courses (MOOCs), with a focus on the use of topic modeling and PyLDAvis visualization to analyze and summarize learner engagement. They used Latent Dirichlet Allocation (LDA) to identify key topics in course content and discussions, which improved their understanding of learner interactions and participation patterns. The study demonstrated how PyLDAvis, an LDA visualization tool, could effectively present topic models, allowing educators to better interpret data and gain insights into student engagement. Onah and Pang used these methodologies to improve the design and delivery of MOOCs, resulting in a better learning experience for participants.

Rajendra Prasad *et al.* (2021) investigated the use of visual topic models to cluster healthcare data. They investigated how these models could improve data analysis in the healthcare sector by producing more understandable and interpretable outcomes. The study used Latent Dirichlet Allocation (LDA) on large datasets to identify hidden patterns and topics within the data. The authors demonstrated that visual topic models improved clustering accuracy over traditional methods. They also emphasized the models' ability to uncover valuable insights in patient records and medical research data. The study included a thorough evaluation of the models' performance, demonstrating their ability to handle complex healthcare information. The authors concluded that visual topic models were an effective tool for analyzing healthcare data, resulting in better decision-making and research outcomes.

Vangara *et al.* (2021) explored the challenge of determining the optimal number of latent topics in a given text corpus using semantic non-negative matrix factorization (SNMF). The authors wanted to improve the accuracy and interpretability of topic modeling by incorporating semantic information into the factorization process. They examined existing methods for topic determination and discovered limitations in their ability to capture semantic coherence. The authors proposed a novel approach to improving topic identification by combining SNMF with semantic coherence measures. They conducted extensive experiments on a variety of text datasets to assess the effectiveness of their method. The results showed that their approach outperformed traditional non-negative matrix factorization (NMF) techniques in terms of extracted topic coherence and relevance. The authors also provided a thorough analysis of how different parameters affected the performance of their model. Their findings indicated that incorporating semantic information into the matrix factorization process significantly improved the quality of topic modeling. The study concluded with a discussion of their method's potential applications

in topics include retrieving information, mining of text, and natural language interpretation, with a focus on its contribution to the advancement of topic modeling techniques.

Alfajri *et al.* (2022) Investigated the use of NMF in analyzing Instagram comments in order to improve Telkom University's entry selection process. The authors aimed to address the challenge of processing large amounts of social media data in order to extract meaningful insights that could be used in university admissions. They collected Instagram comments about Telkom University and used NMF to identify prominent topics within those comments. The analysis identified key themes and sentiments expressed by prospective students and the general public. By organizing comments into coherent topics, the authors demonstrated how NMF could be used to gain a better understanding of applicants' perceptions and concerns. The findings indicated that NMF was an effective tool for summarizing and interpreting unstructured social media data, providing valuable information to the university's admissions committee. Furthermore, the study highlighted the potential of NMF in improving data-driven decision-making processes in higher education settings. The authors' approach helped to advance the field of educational data mining and social media analytics.

Athukorala and Mohotti (2022) explored an innovative approach to topic modeling for short texts on Twitter. The authors addressed the issues raised by the brevity and sparsity of tweets, which frequently impede traditional topic modeling techniques. They proposed a novel method that combines Non-negative Matrix Factorization (NMF) and neighborhood assistance to improve the coherence and relevance of the extracted topics. The authors sought to improve topic detection accuracy by leveraging contextual information from related tweets. They ran extensive experiments on real-world Twitter datasets and found that their method outperformed traditional NMF and Latent Dirichlet Allocation (LDA) models in terms of topic quality and interpretability. The evaluation metrics included coherence scores and human judgment assessments, both of which showed significant improvement. The authors also highlighted their model's scalability, which makes it suitable for processing large amounts of Twitter data. They also discussed the practical applications of their findings, such as social media monitoring and sentiment analysis. Overall, the study provided a solid framework for improving topic modeling in short-text environments, adding important insights to the fields of text mining and natural language processing.

Chu *et al.* (2022) presented a sophisticated model that combines topic modeling and sentiment analysis to identify and visualize themes in literary texts. The authors used Latent Dirichlet Allocation (LDA) topic modeling to identify underlying thematic structures in various literary works. In addition, they used sentiment analysis to determine the emotional tone of the identified topics. The authors tested their model on a wide range of literary texts, demonstrating its effectiveness in detecting nuanced themes and sentiments. They used visualization techniques to represent thematic and sentiment trends, resulting in a more intuitive understanding of the textual data. The study emphasized the model's ability to handle large amounts of text while producing meaningful insights that could be useful for literary analysis and criticism. The authors also discussed how their findings could improve digital humanities research and enable new forms of literary scholarship. The authors made a significant contribution to the field of computational literary studies by developing a comprehensive tool for theme and sentiment analysis, as well as a robust methodology for investigating complex literary phenomena.

Churchill and Singh (2022) analyzed the evolution of the unsupervised topic models from their start till their analysis. Firstly, with the basic introduction of topic modeling they discussed evaluation methods for topic modeling which includes coverage and coherence metrics in which they stated that both these metrics are not enough to judge the topic models so, a qualitative evaluation is also necessary. Secondly, they discussed the early topic models (1990-2006) in which includes Latent Semantic Indexing (LSI) precursor to topic modeling. After that they discussed Latent Dirichlet Allocation (LDA), and the Dirichlet Multinomial Mixture (DMM) the first topic techniques. They also discussed about their methods and LDA variants for different kinds of datasets. After discussion of early model, they gave brief information about time-based and accessible topic models (2006-2011) and then about the modern topic models (2011-Present) which includes Non-negative matrix factorization (NMF) and Graph-based Models. Furthermore, they stated that how traditional topic models can be mixed with modern NLP methods. After briefing some further topic models in the end they gave conclusion which can be stated as by the change of data more and more topic models are used and as data is evaluating further many topic models will be produced and used.

Egger (2022) explored the application of topic modeling to uncover hidden semantic structures in textual data within the tourism industry. The research was presented in a chapter of the book "Applied Data Science in Tourism: Interdisciplinary Approaches, Methodologies, and Applications." Egger utilized advanced topic modeling techniques to

analyze large volumes of text data, aiming to reveal underlying themes and patterns. The study emphasized the importance of these techniques in understanding and interpreting vast amounts of unstructured textual information. By applying these methods, Egger demonstrated how researchers and practitioners could gain valuable insights into tourism-related data. The chapter provided a comprehensive overview of the methodologies, their implementation, and their practical applications in tourism research. Egger's work contributed to the growing field of data science in tourism by showcasing the potential of topic modeling in extracting meaningful information from complex datasets.

Gurcan (2022) analyzed the evolution of bioinformatics research over the past 50 years. Utilizing a comprehensive dataset of 71,490 articles, the authors employed automated text-mining methodologies based on probabilistic topic modeling to uncover the primary topics and trends within the field. They identified 24 major topics that encapsulated the focus and progression of bioinformatics research. The study delineated the temporal evolution of these topics, segmenting the development of the field into seven distinct phases, ranging from the "newborn" to the "wisdom" stages. Significant findings included a recent surge in interest in topics such as "Statistical Estimation," "Data Analysis Tools," "Genomic Data," "Gene Expression," and "Prediction." Furthermore, the analysis revealed a growing emphasis on innovative computing and data analysis methods, particularly those leveraging artificial intelligence and machine learning. This trend underscored the significant advancements in contemporary analysis tools and techniques centered around predictive capabilities. The results provided a holistic overview of bioinformatics research, highlighting the interdisciplinary nature of the field and its rich, evolving literature.

Khan *et al.* (2022) looked into the use of a semantic-based Non-negative Matrix Factorization (NMF) model to predict topic titles from news articles. They wanted to improve the accuracy and efficiency of topic title generation by incorporating semantic information. The study examined various semantic-based approaches and identified the limitations of current models. The authors improved the NMF model by incorporating semantic context to better capture the essence of news articles. They ran extensive experiments to validate their model, demonstrating its superiority to traditional methods. The findings revealed significant improvements in the relevance and coherence of the generated topic titles. This study contributed to the field of automated text summarization and topic modeling by providing a more reliable method for analyzing news articles. The authors concluded that their semantic-based NMF model was an effective solution for automatic topic title prediction.

Nugumanova *et al.* (2022) explored a Non-negative Matrix Factorization (NMF)-based approach to automatic term extraction in their research paper. They wanted to increase the accuracy and efficiency of term extraction processes. The authors compared NMF to other traditional methods and found that it outperformed them in identifying relevant terms. They ran the NMF model on a dataset, extracting terms that were more contextually relevant and meaningful. The study emphasized the benefits of NMF in handling large text corpora and reducing dimensionality. Furthermore, the authors discussed the significance of their findings for natural language processing tasks. They concluded that NMF provided a strong framework for term extraction, significantly outperforming existing techniques. The study contributed to the field by introducing a novel methodology with practical applications in a variety of domains.

Preetham *et al.* (2022) conducted a comparative analysis of Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF) for categorizing research papers into relevant topics. They recognized the importance of efficient topic categorization for research papers, as it facilitates knowledge discovery, information retrieval, and the identification of emerging research trends. The authors assembled a corpus of research papers spanning various disciplines and preprocessed the text data, applying techniques such as tokenization, stop word removal, and stemming/lemmatization to prepare the corpus for topic modeling. They then applied both LDA and NMF topic modeling techniques to the preprocessed corpus, carefully tuning the parameters and leveraging appropriate evaluation metrics to assess the quality and interpretability of the generated topic clusters. The authors compared the performance of LDA and NMF in categorizing research papers using metrics such as topic coherence, topic diversity, and categorization accuracy. They analyzed the strengths and weaknesses of each approach, taking into account the unique characteristics of research papers, such as technical terminology and domain-specific jargon. The authors provided detailed analyses and visualizations to illustrate the effectiveness of LDA and NMF in identifying meaningful topic clusters and categorizing research papers accurately. They discussed the implications of their findings for researchers and practitioners working in the field of academic literature analysis, offering insights and recommendations for selecting the most suitable topic modeling technique based on the specific requirements and objectives of the research paper categorization task.

Kumar *et al.* (2023) investigated the use of NLP for optimizing biometric identification systems. Their research, presented in the 2023 3rd International Conference

on Advance Computing and Innovative Technologies in Engineering (ICACITE), aimed to increase the precision and effectiveness of biometric proof of identity using advanced NLP techniques. They investigated various NLP models and their integration with biometric data to enhance identification processes. The researchers compared the performance of traditional biometric methods to those enhanced with NLP, and found significant improvements in identification accuracy. They also addressed issues related to data privacy and security in NLP-driven biometric systems. The study provided a thorough assessment of the potential benefits and drawbacks of using NLP in biometric identification, providing valuable insights for future research and development in the field. Their findings demonstrated NLP's promising role in the advancement of biometric technologies.

Shastry and Prakash (2023) provided a thorough examination of three common topic modeling techniques: LDA, LSA, and NMF. The authors conducted a orderly comparison of these methods to determine their efficacy in analyzing web data. The study used a diverse dataset sourced from various web domains to ensure robust and generalizable findings. The authors evaluated each method using criteria such as topic coherence, computational efficiency, and the interpretability of generated topics. They discovered that, while LDA produced more coherent topics, it required significant computational resources. LSA, on the other hand, showed faster processing times but produced less interpretable results. NMF achieved a balance of coherence and computational efficiency, making it a viable option for some applications. The authors concluded that the choice of topic modeling technique should be constructed on the research's specific requirements, for example topic coherence, speed, and interpretability. This study provided valued visions into the strengths and limitations of the LDA, LSA, and NMF, assisting researchers in determining the best method for their web data analysis requirements.

Takei *et al.* (2023) explored the themes in higher education research using modeling of topics. The data used in their research consisted of abstracts from journals of educational institutions, research in higher education, and higher education reviews published between 1991 and 2020. The authors extracted 24 main topics from postsecondary research, and the topics that were trending at the time were presented alongside the less popular topics. The methodology included collection of data from higher education journals, data management, analysis, and results. The description about the major themes found in postsecondary research identified the research trends in those years of journals publications, their study gap, and future trend which could be possible after extracting the information from abstracts.

Wang and Zhang (2023) introduced a novel approach to topic modeling by combining NMF with deep learning techniques. Traditional NMF methods assume a linear connection among the input data and the latent topic representations, which may not be sufficient to capture complex and hierarchical topic structures present in many text corpora. To address this limitation, the authors proposed a deep NMF model that learns a non-linear transformation of the input data using a neural network architecture. The deep NMF model consisted of multiple layers, where the first layer performed a non-linear transformation of the input data, and subsequent layers learned higher-level topic representations. The authors developed a novel optimization technique to train the deep NMF model end-to-end, ensuring that the learned topic representations were coherent and interpretable. They evaluated their approach on various text corpora, including news articles, scientific papers, and online reviews. The authors compared the performance of their deep NMF model with traditional NMF, Latent Dirichlet Allocation (LDA), and other state-of-the-art topic modeling methods consuming metrics for example topic coherence, the topic diversity, and the topic quality. The results demonstrated that the deep NMF model outperformed existing techniques, uncovering more coherent and interpretable topics that better captured the underlying semantic structures present in the text data. The authors provided detailed analyses and visualizations to illustrate the effectiveness of their approach in identifying hierarchical and complex topic structures, which were difficult to uncover using traditional topic modeling methods.

Amaro and Bacao (2024) proposed a consistent framework for conducting comparative studies of topic modeling techniques. They recognized the challenges in comparing different topic modeling algorithms due to the lack of standardized evaluation metrics and experimental setups. The authors developed a comprehensive framework that included guidelines for data preprocessing, parameter tuning, and evaluation. They emphasized the importance of consistent data preprocessing steps, such as tokenization, stop word removal, and stemming/lemmatization, to ensure fair comparisons across different algorithms. The authors also discussed strategies for parameter tuning, highlighting the need to explore a range of parameter values and select the optimal settings based on appropriate evaluation metrics. They introduced a set of evaluation metrics, including topic coherence, topic diversity, and topic quality, and provided guidelines for interpreting and analyzing the results. The authors demonstrated the application of their framework by conducting a comparative study of several topic modeling techniques, including LDA, NMF, and Correlated Topic Model (CTM), on various text corpora. They

presented detailed analyses and visualizations to illustrate the strong point and flaws of each algorithm, and provided recommendations for selecting the most suitable technique based on the characteristics of the text data and the specific research objectives.

Kumar and Gupta (2024) investigated how data science and analytics could be combined with modeling and simulation in the context of omics datasets. Their research addressed a variety of issues related to managing and analyzing large-scale omics data. The authors emphasized the need for advanced analytical techniques to extract meaningful insights from complex biological data. They talked about the challenges of data preprocessing, integration, and interpretation for omics datasets. Furthermore, the study emphasized the importance of simulation models in understanding biological processes and systems. Kumar and Gupta examined the current methodologies and tools used in omics research, identifying limitations and potential improvements. They also proposed methods to improve the reliability and accuracy of data analysis in this field. The paper emphasized the importance of interdisciplinary approaches to addressing the complexities of omics data. Overall, their findings provided valuable insights into the convergence of data science and biology in omics research.

Nanyonga *et al.* (2024) shepherded a relative analysis of various topic modeling methods on text narratives from the Australian Transport Safety Bureau (ATSB). They recognized the importance of extracting meaningful insights from these narratives, which often contain rich information about transportation incidents and accidents. The authors preprocessed the ATSB text narratives with methods that use natural language processing such as the use of tokens stopword removal, and the lemmatization. They then used multiple topic modeling methods on the preprocessed data, including LDA, NMF, and Biterm Topic Model (BTM). The authors measured the effectiveness of various algorithms using parameters such as topic coherence, variation, and interpretation. They compared the generated topics across different algorithms and analyzed their strengths and weaknesses in capturing relevant information from the ATSB narratives. The authors provided detailed visualizations and qualitative analyses to illustrate the effectiveness of each technique in identifying meaningful topics related to transportation incidents and accidents. They discussed the implications of their findings for practitioners and researchers working in the field of transportation safety analysis, offering insights and recommendations for selecting the most appropriate topic modeling technique based on the characteristics of the text data and the specific analysis objectives.

Moreno-Ortiz (2024) explored the concept of "topics" in the context of topic modeling and text inquiry. They delivered a comprehensive impression of the various definitions and interpretations of topics, as well as the different approaches to representing and extracting topics from text data. The authors discussed the historical development of topic modeling techniques, starting from early methods like LSA and progressing to more recent algorithms such as LDA and NMF. They examined the underlying assumptions and mathematical foundations of these techniques, highlighting their strengths and limitations in capturing topics from text data. The authors also explored the challenges and considerations involved in evaluating and interpreting the generated topics, including the role of human interpretation and domain expertise. They presented case studies and examples after numerous domains, such for example news apprenticeships, scientific literature, and social media data, to illustrate the practical applications and impact of topic modeling. The authors discussed the emerging trends and future directions in topic modeling research, including the incorporation of deep learning methods and the investigation of novel evaluation metrics. They also addressed the ethical thoughts and potential biases related with topic modeling, emphasizing the importance of responsible and transparent use of these techniques in real-world applications.

Muthusami *et al.* (2024) conducted a comprehensive evaluation of various topic modeling techniques, focusing on their performance in capturing meaningful topics from short-text data across diverse domains. They noticed the difficulties presented by short texts, such as social networking postings, product reviews, and headlines from the news, where classic topic modeling methods frequently fail because of a lack of significant word overlap evidence. The authors investigated a variety of topic-specific modeling strategies, including LDA, NMF, and other cutting-edge methods created expressly for short-text data. They compiled a varied corpus of short-text datasets from many areas, including news, social media, and e-commerce. The authors preprocessed the datasets, applying appropriate techniques to handle the unique characteristics of short texts, such as abbreviations and informal language. They then applied each of the selected topic modeling techniques to the preprocessed datasets, carefully tuning the parameters and leveraging appropriate evaluation metrics to assess the quality and interpretability of the generated topics. The authors compared the performance of the various techniques across different domains using metrics such as topic coherence, topic diversity, and topic quality. They analyzed the strengths and weaknesses of each approach in capturing meaningful topics from short-text data, taking into account the specific characteristics and challenges of each domain. The

authors provided detailed analyses and visualizations to illustrate the effectiveness of different topic modeling techniques in different contexts, offering insights and recommendations for researchers and practitioners working with short-text data.

Pavithra and Savitha (2024) conducted a comprehensive study to investigate the performance of various topic modeling techniques for analyzing evolving textual data, with a specific focus on research papers. They compared the effectiveness of LDA, Hierarchical Dirichlet Process (HDP), NMF, BERTOPIC, and Dynamic Topic Model (DTM) in capturing the temporal dynamics and topic evolution present in a corpus of research papers spanning multiple years. The authors first preprocessed the textual data, which involved steps such as tokenization, stop word removal, and stemming, to prepare the corpus for topic modeling. They then applied each of the aforementioned algorithms to the preprocessed data, carefully tuning the parameters and leveraging appropriate assessment metrics to measure the excellence and interpretability of the generated topics. The authors found that DTM and BERTOPIC excelled in capturing the sequential development of topics, effectively identifying emerging and declining themes over time. On the other hand, LDA and NMF demonstrated superior performance in identifying stable and interpretable topics that remained consistent throughout the corpus. The authors provided detailed analyses and visualizations to illustrate the strengths and weaknesses of each approach, highlighting the trade-offs between capturing topic dynamics and ensuring topic coherence and interpretability. They concluded that the choice of the most suitable topic modeling algorithm depends on the specific research objectives and the characteristics of the textual data under investigation.

CHAPTER 3

MATERIALS AND METHODS

In this chapter, the methodology used to investigate recent bioinformatics research trends using PubMed data for topic modeling is described. First, how to download abstracts and topic names from research articles published in PubMed, as well as the criteria used to ensure relevance to bioinformatics is described. Next, the preprocessing pipeline, which includes lemmatization, stopwords removal, tokenization, stemming, and part-of-speech tagging is described. This preprocessing is critical in preparing the textual data for analysis. In this research it is explained why each preprocessing step is necessary and what tools are used for implementation. Finally, the use of the LDA and NMF algorithm for topic modeling is discussed, emphasizing its importance in revealing hidden themes within a corpus of abstracts. Through this detailed methodology, insights into the complexities of data preparation and analysis for elucidating bioinformatics research trends is provided, laying the groundwork for more in-depth discussions in later chapters. The planned methodology for the text pre-processing is given in figure.

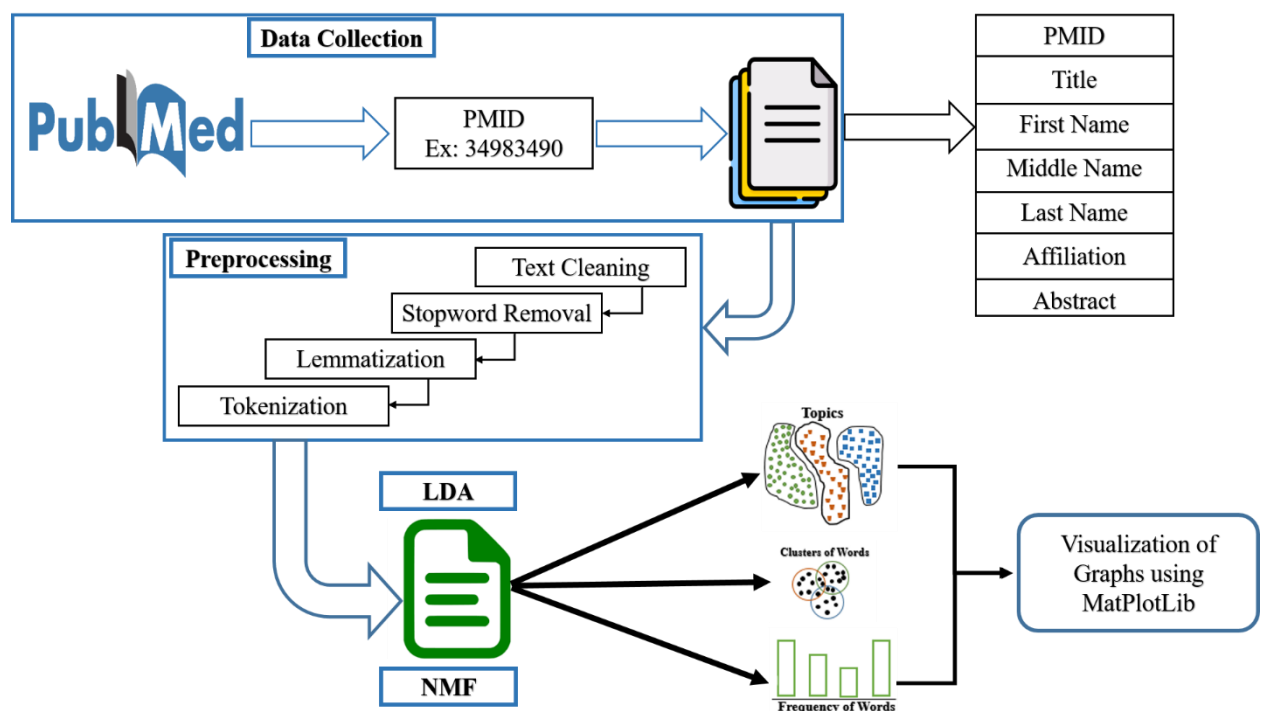


Figure 3.1 Proposed methodology for the research trend ideation in bioinformatics

3.1 Data collection

First step in this study is collecting the data relevant to the bioinformatics research which is going to be used for topic modeling. By identifying the set of keywords related to the field of bioinformatics search query was placed on PubMed database which included the

keywords 'Docking', 'Molecular Docking', 'Molecular Dynamic Simulation', 'Phylogenetic Analysis', 'Single Nucleotide Polymorphism', 'Gene expression', 'Next generation sequence analysis', 'NGS Analysis' 'In- silico bioinformatics Analysis', 'Bioinformatics analysis', 'Computational analysis', 'In-silico drug discovery', 'Drug discovery', 'In-silico structural analysis', 'Structural analysis', 'In-silico functional analysis', 'Functional analysis', 'Mutational analysis', 'Virtual screening', 'Sequence annotation' 'Genome sequence', 'analysis', 'Proteome sequence analysis', and 'RNA-seq analysis' which gave me search results on PubMed webpage. After that some filters were applied to ensure the relevance of the obtained data according to my research directions. Those filters included the time filter which was used to gather the research articles for the last three years, one was for those research articles in which abstracts were available because the objective of this study was to perform topic modeling on abstracts of the research articles, and the other filter used in this study was to check which articles were free to download so that the abstracts and topic names of the articles can be retrieved. After retrieving the articles, essential metadata fields such as PubMed ID (PMID), abstracts, topic names, journal information, and publication dates, resulting in a comprehensive dataset for subsequent analysis were meticulously extracted. This information was given in a txt file by the PubMed which contained some extra metadata which was none of our use for example owner information, Copyright information, Address, Title Abbreviation, Medical Subject Headings, Conflict of Interest Statement, Publication History Status Tags, Publication Status etc.

3.2 Dataset Preparation

This research endeavored to construct a comprehensive dataset culled from the extensive PubMed database, a preeminent repository of biomedical literature curated by the National Library of Medicine. The overarching objective was to distill and consolidate pivotal metadata elements associated with scientific publications, thereby facilitating subsequent analyses and explorations within the realm of bioinformatics research.

In the inaugural phase, the requisite data from PubMed, a database replete with a vast assemblage of metadata fields corresponding to each indexed publication was procured. This raw dataset, teeming with an array of metadata attributes, laid the foundation for the ensuing data processing and refinement stages. To sculpt the dataset into a more concentrated and purposeful form, a Python script was meticulously crafted to extract and isolate specific metadata fields deemed essential for the study's objectives. These fields

encompassed the PubMed Identifier (PMID), a unique identifier assigned to each publication, the Title, a succinct encapsulation of the paper's subject matter, and the Abstract, a concise yet informative synopsis of the research findings and methodologies employed. Additionally, the script captured the Year of publication, a temporal marker pivotal for chronological analyses and trend identification. The Place of Publication and Journal Title fields were also extracted, providing contextual information regarding the dissemination channel and the esteemed scholarly journal in which the research was promulgated.

```

1 import re
2 import csv
3 import os

4 # Regular expression patterns for different fields
5 pmid_pattern = re.compile(r'PMID- (\d+)', re.IGNORECASE)
6 ti_pattern = re.compile(r'TI - (.+?)(?=PG -|LID -|DP -|LA -|PT -|DEP -|PL -|JT -|JID -|\Z)', re.DOTALL)
7 abstract_pattern = re.compile(r'AB - (.+?)(?=(?:CI -|FAU -|FAU -|DP -|TI -|\Z))', re.DOTALL)
8 dp_pattern = re.compile(r'DP - (.+?)(?=TI -|\Z)', re.DOTALL)
9 pt_pattern = re.compile(r'PT - (.+)$', re.MULTILINE)
10 pl_pattern = re.compile(r'PL - (.+?)(?=TA -|\Z)', re.DOTALL)
11 jt_pattern = re.compile(r'JT - (.+)$', re.MULTILINE)

12 # Function to extract data from a single file
13 def extract_data_from_file(file_path):
14     with open(file_path, 'r', encoding='utf-8') as file:
15         text = file.read()
16
17     # Find all PMIDs
18     pmids = pmid_pattern.findall(text)
19
20     # Extract data using the list of PMIDs and regular expressions
21     data = []
22     for pmid in pmids:
23         record_pattern = re.compile(rf'PMID- {pmid}(.*?)\n\n', re.DOTALL)
24         record_match = record_pattern.search(text)
25
26         if record_match:
27             record_text = record_match.group(1)
28
29             abstract_match = abstract_pattern.search(record_text)
30             dp_match = dp_pattern.search(record_text)
31             ti_match = ti_pattern.search(record_text)
32             pt_match = pt_pattern.search(record_text)
33             pl_match = pl_pattern.search(record_text)
34             jt_match = jt_pattern.search(record_text)

```

```

33
34     abstract = abstract_match.group(1).strip() if abstract_match else ''
35     dp = dp_match.group(1).strip() if dp_match else ''
36     ti = ti_match.group(1).strip() if ti_match else ''
37     pt = pt_match.group(1).strip() if pt_match else ''
38     pl = pl_match.group(1).strip() if pl_match else ''
39     jt = jt_match.group(1).strip() if jt_match else ''
40
41     data.append([pmid, ti, abstract, dp, pt, pl, jt])
42
43     return data
44
45 # List of files to process
46 file_list = [
47     '1.txt', '1.txt', '2.txt', '3.txt', '4.txt', '5.txt',
48     '6.txt', '7.txt', '8.txt', '9.txt', '10.txt',
49     '11.txt', '12.txt', '13.txt', '14.txt', '15.txt',
50     '16.txt', '17.txt', '18.txt', '19.txt', '20.txt',
51     '21.txt', '22.txt'
52 ]
53
54 # Aggregate data from all files
55 all_data = []
56 for file_name in file_list:
57     if os.path.isfile(file_name):
58         file_data = extract_data_from_file(file_name)
59         all_data.extend(file_data)
60
61 # Writing the aggregated data to a single CSV file
62 with open('output.csv', 'w', newline='', encoding='utf-8') as file:
63     writer = csv.writer(file)
64     writer.writerow(['PMID', 'TITLE', 'ABSTRACT', 'DATE OF PUBLICATION', 'PUBLICATION TYPE', 'PLACE OF PUBLICATION', 'JOURNAL'])
65     writer.writerows(all_data)
66
67 print("Extraction complete. Output saved to output.csv")
68

```

Extraction complete. Output saved to output.csv

Figure 3.2 Data Extraction from PubMed text

3.2.1 Data Cleaning and Normalization for Dataset

Subsequent to the extraction phase, a series of data cleaning and normalization procedures were implemented to enhance the dataset's integrity, consistency, and usability. These measures included:

1. Removal of extraneous spaces: The script diligently identified and eliminated any superfluous spaces between words within the extracted metadata fields, thereby promoting a streamlined and uniform presentation of the dataset.
2. Elimination of redundant new line characters: Akin to the treatment of excess spaces, the script systematically purged any unnecessary new line characters ("\n") from the metadata fields, ensuring a coherent and uninterrupted flow of information.
3. Year isolation: Recognizing the variability in date formatting across publications, with some entries including the month and day while others omitted these granular details, the year component as a dedicated field was extracted. This measure not only promoted consistency but also facilitated any temporal analyses or filtration processes predicated on the publication year.

```

1 import pandas as pd
2 import re

3
4 # Load the data
5 df = pd.read_csv('file_name')
6
7 # Define a function to extract the first four digits (year) from a string
8 def extract_year(date_string):
9     match = re.search(r'\b(\d{4})\b', date_string)
10    return match.group(1) if match else None
11
12 # Apply the function to the 'DATE OF PUBLICATION' column
13 df['DATE OF PUBLICATION'] = df['DATE OF PUBLICATION'].apply(extract_year)
14
15
16 def clean_text(text):
17     if isinstance(text, float):
18         return "" # Return an empty string for non-string values
19     text = text.replace('\n', ' ') # Replace newlines with space
20     text = text.replace('\xa0', ' ') # Replace non-breaking space with space
21     text = ' '.join(text.split()) # Remove extra spaces
22     return text
23
24 # Apply the cleaning function to the 'TITLE_ABSTRACT' column
25 df['ABSTRACT'] = df['ABSTRACT'].apply(clean_text)
26 df['TITLE'] = df['TITLE'].apply(clean_text)
27 # Save the cleaned DataFrame to a new CSV file
28 df.to_csv('cleaned_output.csv', index=False)
29 print("Year extraction complete. Output saved to cleaned_output.csv")

```

Year extraction complete. Output saved to cleaned_output.csv

Figure 3.3 Year, Title and Abstract cleaning

By adhering to these rigorous data preparation protocols, the study has yielded a meticulously curated dataset encompassing essential metadata elements culled from PubMed publications. This dataset, comprising PMID, Title, Abstract, Year, Place of Publication, and Journal Title, has been systematically cleaned and normalized, rendering it an invaluable resource for further investigations and inquiries within the realms of biomedical literature and scientific discourse. The cleaned and standardized dataset was then saved as a CSV file for further analysis and research purposes.

3.3 Data Pre-processing

In this step, the collected data was preprocessed to prepare it for LDA and NMF algorithms. The preprocessing techniques used on dataset are given in following figure.

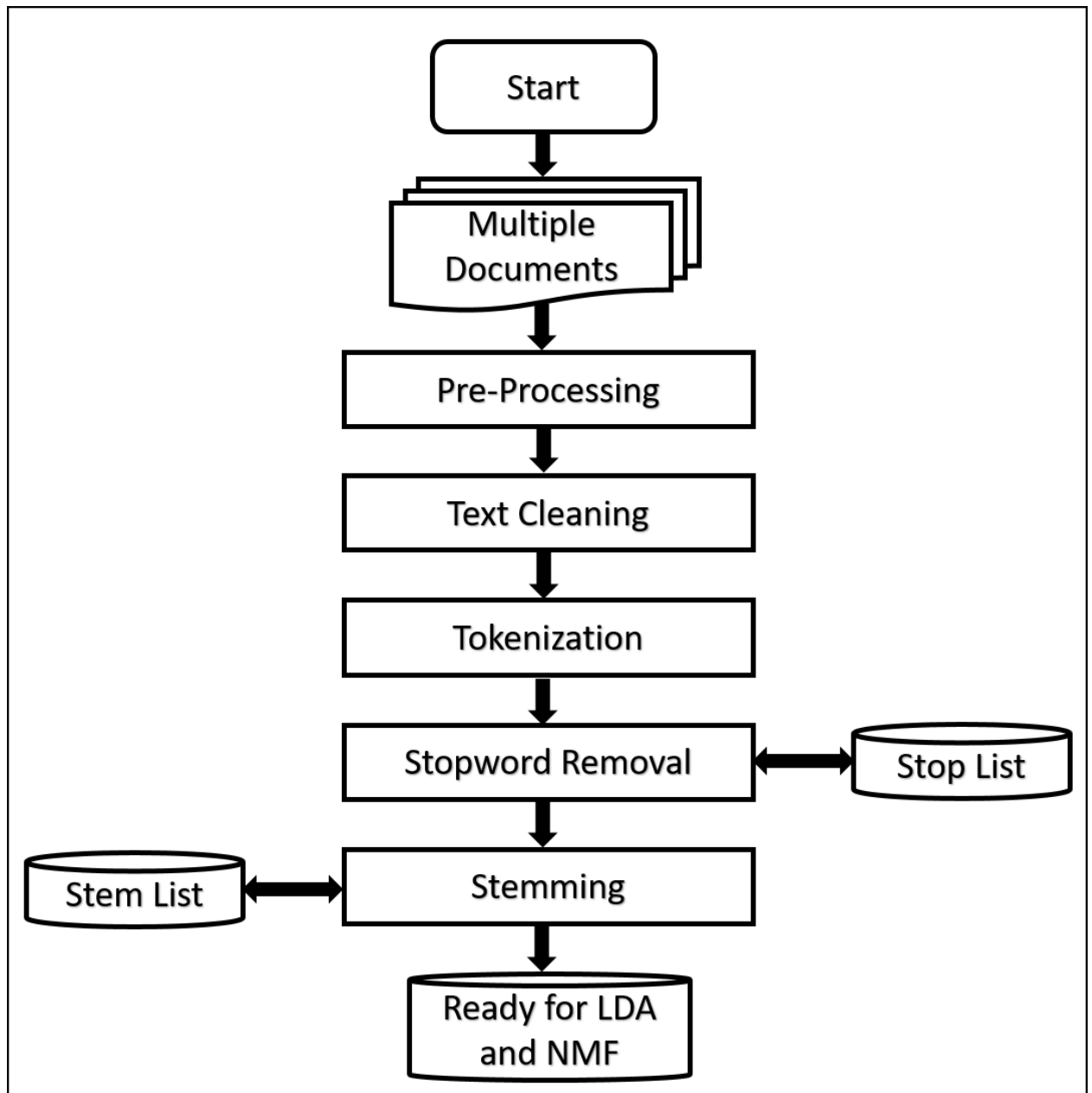


Figure 3.4 The preprocessing techniques

3.3.1 Dropping Duplicates and Null/Empty Values

This study took steps to clean and preprocess the dataset by dropping duplicate entries and removing rows with null or empty values. Duplicate entries can introduce redundancy and bias in the analysis, while null or empty values represent missing data that can lead to inaccurate or incomplete results. Specifically, the code implemented functionality to identify and remove exact duplicate rows from the dataset, ensuring that each entry was unique and no redundant information was present. Additionally, the study handled null or empty values by either dropping the entire row containing such values or employing

appropriate imputation techniques to estimate and fill in the missing values based on the available data.

By eliminating duplicate entries and rows with null or empty values, this study aimed to enhance the quality, integrity, and reliability of the dataset before proceeding with the topic modeling analysis using algorithms like LDA and NMF. This preprocessing step contributed to the overall robustness and validity of the research findings by ensuring that the analysis was performed on clean, unique, and complete data.

Dropping Duplicates and Null/Empty Values ¶

```
1 # Specify the columns you want to check for NaN and duplicates
2 columns_to_check = ['TITLE', 'ABSTRACT']
3
4 # Drop Null/Epty values based on the specified columns
5 df = df.dropna(subset=columns_to_check)
6
7 # Drop duplicates based on the specified columns
8 df = df.drop_duplicates(subset=columns_to_check)
```

Figure 3.5 Dropping Duplicates and Null/Empty Values

3.3.2 Lowercasing:

Lowercasing is a crucial preprocessing step in which all text data was converted to lowercase to ensure uniformity and consistency throughout the document corpus. By doing so, the model was prevented from treating the same word differently based on its casing, thus improving the accuracy of subsequent analyses. For example, without lowercasing, "Word" and "word" would be considered as two separate entities, potentially leading to fragmented or inaccurate topic modeling results. By applying lowercasing, the text data was standardized, enabling more effective tokenization, stopword removal, and stemming or lemmatization processes downstream in the preprocessing pipeline. This normalization step ensures that the subsequent algorithms, such as LDA and NMF, operate on a clean and consistent input, facilitating more accurate and meaningful topic extraction from PubMed abstracts.

3.3.3 Stopword Removal:

In the preprocessing stage, stopwords removal is a crucial step aimed at eliminating common words that do not carry significant semantic meaning in the context of topic modeling. This technique was employed by leveraging libraries such as NLTK to filter out words like "the," "and," "is," and so forth from the abstracts extracted from PubMed. These stopwords, though ubiquitous in natural language, can skew the results of topic modeling algorithms such as LDA and NMF by adding noise and reducing the discriminative power of the model. By removing them, it was ensured that the focus remains on content-rich terms,

allowing the algorithms to identify meaningful patterns and topics within the dataset accurately. In the following figure stopwords are given which were removed from dataset.

Stopwords

```
1 stopwords = stopwords.words("english")
1 print (stopwords)

['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've", "you'll", "you'd", 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', "she's", 'her', 'hers', 'herself', 'it', "it's", 'its', 'its', 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', "that'll", 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doin', 'g', 'a', 'an', 'the', 'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'o', 'ut', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'al', 'l', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', "don't", 'should', "should've", 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', "aren't", 'couldn', "couldn't", 'didn', "didn't", 'doesn', "doesn't", 'hadn', "hadn't", 'has', 'n', "hasn't", 'haven', "haven't", 'isn', "isn't", 'ma', 'mightn', "mightn't", 'mustn', "mustn't", 'needn', "needn't", 'sha', 'n', "shan't", 'shouldn', "shouldn't", 'wasn', "wasn't", 'weren', "weren't", 'won', "won't", 'wouldn', "wouldn't"]
```

Figure 3.6 Stopwords

3.3.4 Customized Stop Words

Customized stopwords were also employed in the preprocessing step to ensure that domain-specific or relevant terms inadvertently remove from the corpus. Stopwords are commonly used words (e.g., "the," "and," "is") that are typically filtered out during text preprocessing as they carry little semantic meaning. However, in certain domains or contexts, some of these commonly used words may hold significant relevance and should be retained.

To address this, the code allowed for the incorporation of a customized stopwords list. By customizing the stopwords list, this study ensured that potentially important terms specific to the research domain were not inadvertently discarded during the preprocessing phase, thereby retaining valuable context and meaning in the corpus before applying topic modeling techniques like LDA and NMF. Following is the implementation of those customized stopwords.

```
# Remove stopwords
custom_stopwords = ['BACKGROUND', 'MATERIAL AND METHODS', 'RESULTS', 'CONCLUSIONS', 'MOTIVATION', 'IMPLEMENTATION', 'SUPPLEMENTARY INFORMATION', 'OBJECTIVES', ',', ':', 'identified', 'cell', 'cells', 'protein', 'proteins', 'network', 'study', 'virus', 'degs', 'pathways', 'differentially', 'human', 'methods', 'available', 'model', 'new', 'models', 'method', 'using', 'information', 'results', 'data', 'samples', 'species', 'stress', 'development', 'survival', 'variants', 'compounds', 'binding', 'inhibitors', 'potential', 'activity', 'clinical', 'treatment', 'care', 'images', 'hub', 'pathway', 'supplementary', 'information', 'tools', 'reads', 'cnc', 'prognosis', 'lung', 'variant', 'genetic', 'pathogenic', '3d', 'imaging', 'reality', 'vr', 'ct', 'singlecell', 'proliferation', 'signaling', 'spike', 'protease', 'mpro', 'viral', 'sarscov2', 'strain', 'strains', 'noncoding', 'target', 'targets', 'design', 'used', 'based', 'datasets', 'approach', 'prostate', 'colorectal', 'gastric', 'cancers', 'ovarian', 'compound', 'showed', 'vitro', 'peptides']

stop_words = set(stopwords.words('english'))
stop_words.update(custom_stopwords)
data['text'] = data['text'].apply(lambda x: ' '.join([word for word in word_tokenize(x) if word not in stop_words]))
```

Figure 3.7 Customised Stop Words

3.3.5 Stemming or Lemmatization:

In the preprocessing stage of this research, stemming or lemmatization plays a crucial role in standardizing the text data extracted from PubMed abstracts. Stemming involves

reducing words to their base or root form by removing suffixes, which helps in consolidating variations of words into a common form. For instance, words like "running" and "runs" would be reduced to their stem "run." On the other hand, lemmatization involves mapping words to their dictionary form or lemma, considering the word's part of speech. This process ensures that words are transformed into their canonical form, aiding in better analysis and interpretation of the text data. Both stemming and lemmatization contribute to the normalization of the vocabulary, thereby enhancing the effectiveness of topic modeling algorithms like LDA and NMF in identifying meaningful patterns and topics within the abstracts. Following is the lemmatization process done in python.

Lemmatization

```
1 def lemmatization(texts, allowed_postags=["NOUN", "ADJ", "VERB", "ADV"]):
2     nlp = spacy.load("en_core_web_sm", disable=["parser", "ner"])
3     texts_out = []
4     for text in texts:
5         doc = nlp(text)
6         new_text = []
7         for token in doc:
8             if token.pos_ in allowed_postags:
9                 new_text.append(token.lemma_)
10        final = " ".join(new_text)
11        texts_out.append(final)
12    return (texts_out)
13
14
15 lemmatized_texts = lemmatization(data)
16
```

Figure 3.8 Stemming or Lemmatization

3.3.6 Removing Special Characters and Numbers:

In removing special characters and numbers from the text data, a straightforward process to ensure the dataset is clean and focused on meaningful content was executed. Initially, iterations were run through each document in the dataset, examining every word or token within. For each token, it was checked if dataset contains any special characters or numerical digits using regular expressions. If a token includes any such characters, it will be excluded from the final dataset. This process effectively filters out noise that could disrupt the coherence of topics identified by the algorithms. Once completed, the dataset consisted solely of words and phrases relevant to the research topics, facilitating more accurate analysis and interpretation by the LDA and NMF algorithms.

3.3.7 Tokenization:

Tokenization is the process of breaking down a piece of text, such as a sentence or a paragraph, into individual tokens, which can be words, phrases, or symbols. In this research methodology, tokenization plays a crucial role in preparing the PubMed abstracts combined with titles of the research articles for analysis using LDA and NMF algorithms. By

tokenizing the abstracts and topics, transforming them from continuous strings of text into discrete units, allowing for easier manipulation and analysis. For example, the sentence "The quick brown fox jumps over the lazy dog" would be tokenized into individual words: ["The", "quick", "brown", "fox", "jumps", "over", "the", "lazy", "dog"]. This process facilitates subsequent steps such as removing stopwords, stemming or lemmatization, and ultimately, topic modeling. Following figure gives the implementation of tokenization.

Tokenization

```
1 def gen_words(texts):
2     final = []
3     for text in texts:
4         new = gensim.utils.simple_preprocess(text, deacc=True)
5         final.append(new)
6     return (final)
7
8 data_words = gen_words(lemmatized_texts)
9
10 print (data_words[0][0:20])
```

['name', 'bear', 'small', 'town', 'call', 'bear', 'very', 'hard', 'work', 'child', 'father', 'mother', 'small', 'mill', 'flour', 'buckwheat', 'prosperous', 'comfortable', 'go', 'school']

Figure 3.9 Tokenization

3.3.8 Bigrams and Trigrams

In the context of topic modeling, this study recognized the importance of capturing multi-word expressions, known as n-grams, to enhance the accuracy and interpretability of the generated topics. Specifically, the use of bigrams (two-word phrases) and trigrams (three-word phrases) was employed to preserve the contextual meaning of frequently occurring word combinations. This study utilized the gensim library in Python to extract bigrams and trigrams from the preprocessed corpus of documents. The code first identified frequently occurring bigrams and trigrams based on predefined thresholds using the gensim.models.Phrases function. It then created bigram and trigram objects from these identified phrases using gensim.models.phrases.Phaser. Subsequently, custom functions make_bigrams and make_trigrams are defined to transform individual documents by replacing separate words with the detected bigrams and trigrams, respectively. By incorporating bigrams and trigrams into the corpus, this research aimed to capture the contextual meaning of frequently occurring word combinations, enhancing the quality and interpretability of the topics generated by the LDA and NMF algorithms. This process helped to preserve the semantic relationships between words, which can be lost when treating each word independently.

BIGRAMS AND TRIGRAMS

```
1 bigram_phrases = gensim.models.Phrases(data_words, min_count=5, threshold=100)
2 trigram_phrases = gensim.models.Phrases(bigram_phrases[data_words], threshold=100)
3
4 bigram = gensim.models.phrases.Phraser(bigram_phrases)
5 trigram = gensim.models.phrases.Phraser(trigram_phrases)
6
7 def make_bigrams(texts):
8     return([bigram[doc] for doc in texts])
9
10 def make_trigrams(texts):
11     return ([trigram[bigram[doc]] for doc in texts])
12
13 data_bigrams = make_bigrams(data_words)
14 data_bigrams_trigrams = make_trigrams(data_bigrams)
15
16 print (data_bigrams_trigrams[0][0:20])
```

['name', 'bear', 'small', 'town', 'call', 'bear', 'very', 'hard', 'work', 'child', 'father', 'mother', 'small', 'mill', 'flo
ur', 'buckwheat', 'prosperous', 'comfortable', 'go', 'school']

Figure 3.10 Bigrams and Trigrams

3.3.9 Term Frequency-Inverse Document Frequency (TF-IDF)

Finally, to improve retrieval of data based on keyword significance and frequency, preprocessed text data was converted into TF-IDF (Term Frequency-Inverse Document Frequency) weights. The TF-IDF weighs each term in the source material based on how often it's seen (term frequency) and how unusual it is over the full corpus (inverse document frequency). This transformation helps highlight phrases that are significant within a given text but less common throughout the overall dataset, increasing the relevance of keywords for subsequent analysis. Furthermore, TF-IDF recognizes the significance of numbers, ensuring that they contribute meaningfully to document representation while accounting for their varying frequencies. This step refines the data for efficient information retrieval and more accurate topic modeling using LDA and NMF algorithms.

TF-IDF

```
1 from gensim.models import TfidfModel
2
3 id2word = corpora.Dictionary(data_bigrams_trigrams)
4
5 texts = data_bigrams_trigrams
6
7 corpus = [id2word.doc2bow(text) for text in texts]
8 # print (corpus[0][0:20])
9
10 tfidf = TfidfModel(corpus, id2word=id2word)
11
12 low_value = 0.03
13 words = []
14 words_missing_in_tfidf = []
15 for i in range(0, len(corpus)):
16     bow = corpus[i]
17     low_value_words = [] #reinitialize to be safe. You can skip this.
18     tfidf_ids = [id for id, value in tfidf[bow]]
19     bow_ids = [id for id, value in bow]
20     low_value_words = [id for id, value in tfidf[bow] if value < low_value]
21     drops = low_value_words + words_missing_in_tfidf
22     for item in drops:
23         words.append(id2word[item])
24     words_missing_in_tfidf = [id for id in bow_ids if id not in tfidf_ids]#The words with tf-idf score 0 will be missing
25
26     new_bow = [b for b in bow if b[0] not in low_value_words and b[0] not in words_missing_in_tfidf]
27     corpus[i] = new_bow
28
```

Figure 3.11 TF-IDF

3.4 Implementation of Topic Models

After preprocessing, both LDA and NMF algorithms were applied to identify latent topics within the PubMed data.

3.4.1 Model 1: Latent Dirichlet Allocation (LDA)

In this research, LDA as the primary topic modeling technique was employed, acknowledging its widespread use and effectiveness in analyzing text corpus (Blair *et al.*, 2020). LDA operates as a generative probabilistic model within a three-level hierarchical Bayesian framework. Every document within the corpora is represented as a mix over a set of fundamental topics, while for each topic is represented as a mixture over a collection of topic possibilities (Pyo *et al.*, 2015). One significant benefit of LDA is its capacity to automatically determine the sum of topics, removing the necessity for predefinition.

To determine optimal standards for the three hyper parameters essential for LDA – the amount of topics (K), beta, and alpha – I conducted a grid search. Beta influences the word composition of topics, with higher values resulting in topics consisting of more words. Similarly, alpha influences the diversity of topics, with higher values leading to more varied topics. The grid search began with a range of topics from two to 15, with incremental steps

of one. Initially, K was pre-defined, and then the search for beta and alpha values was conducted accordingly.

LDA Model

```
1 lda_model = gensim.models.ldamodel.LdaModel(corpus=corpus[:-1],
2                                             id2word=id2word,
3                                             num_topics=10,
4                                             random_state=100,
5                                             update_every=1,
6                                             chunksize=100,
7                                             passes=10,
8                                             alpha="auto")
9
```

Figure 3.12 LDA Model

During the grid search process, one hyperparameter was maintained constant while varying the other until achieving the highest coherence score, which assesses the quality of the extracted topics. Ultimately, a coherence score of 0.48 was attained for 14 topics. Subsequently, the grid search identified symmetric distributions for both alpha and beta, with values of 0.91 each, indicating well-balanced topic compositions.

To aid in the explanation of the extracted evidence from the fixed LDA topic model, pyLDavis was utilized to make an interactive distance map, facilitating a clearer understanding of the topic relationships within the corpus.

3.4.2 Model 2: Non-negative Matrix Factorization (NMF)

NMF, unlike LDA, is a decompositional, non-probabilistic algorithm that utilizes matrix factorization techniques, falling under the category of linear-algebraic algorithms (Egger, 2022). The process begins with TF-IDF transformed data, where each term's importance across a gathering of papers is evaluated. As depicted in Figure 2, NMF breaks down a term-document matrix (A) into two lower-ranking matrices: a terms-topics matrix (W) and a topics-documents matrix (H) (Chen *et al.*, 2019). The values of W and H are changed iteratively, with W carrying the basis variables and H containing the associated weights. To facilitate topic interpretation, all entries in W and H must be non-negative in nature as negative values could complicate the process.

```

70 # Fit NMF model: See [1] for more details
71 nmf_model = gensim.models.Nmf(
72     corpus = corpus,      # Document-Term Matrix
73     id2word = id2word,    # Map word IDs to words
74     num_topics = 10,      # Number of latent topics to extract
75     random_state = 100,
76     passes = 100,         # N° of passes through the corpus during training
77 )
78
79 # Get the topics sorted by sparsity
80 nmf_model.show_topics()

```

Figure 3.13 NMF Model

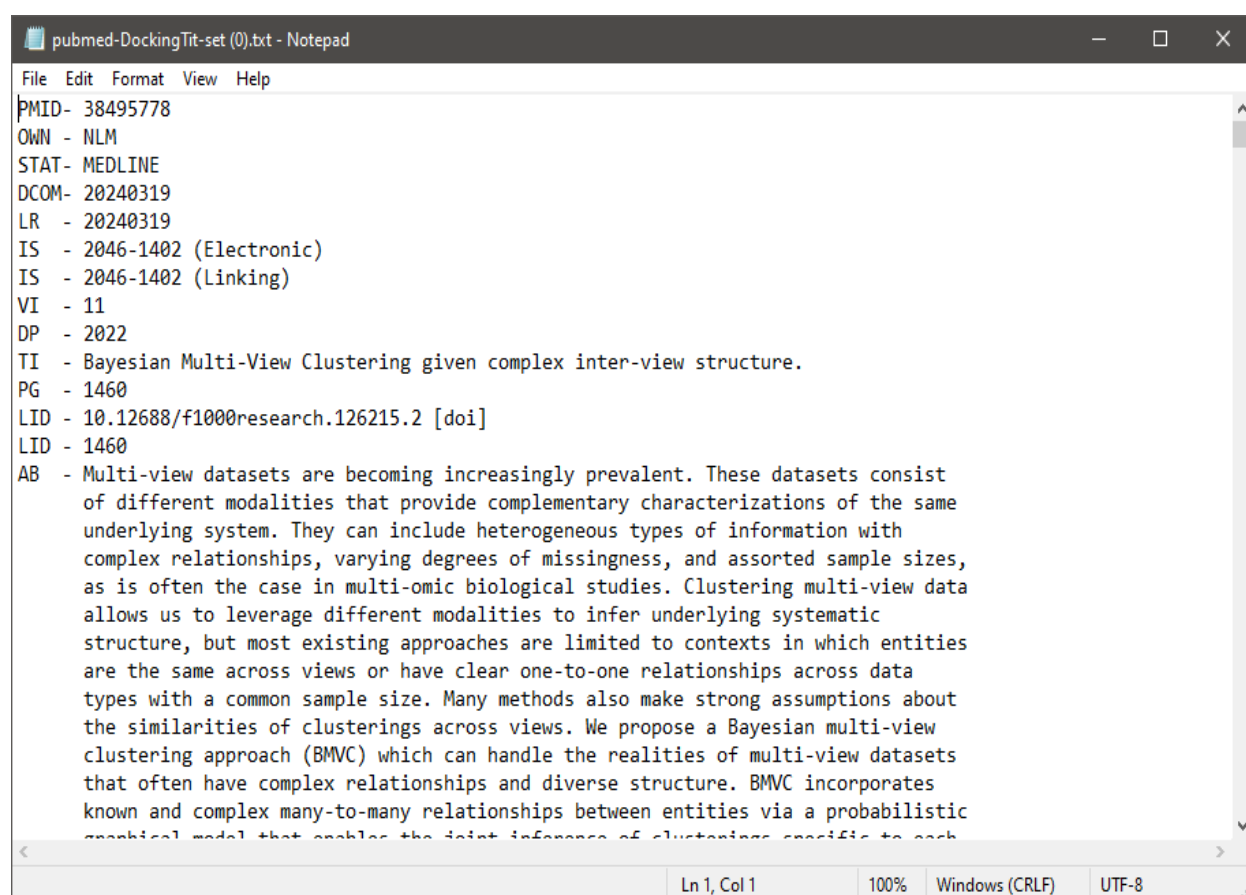
Given NMF's requirement for preprocessed data, it necessitates classical NLP pipeline steps such as lowercasing, stopword removal, lemmatizing or stemming, as well as punctuation and number elimination (Egger, 2022). In this study, the open-source Python library Gensim was utilized to determine the optimal number of topics. By computing the highest coherence score, this study was able to identify 10 distinct topics within the corpus.

CHAPTER NO 4

RESULTS AND DISCUSSION

4.1 Raw Data

In this section results are given. Firstly, the data got after searching query and applying filters on PubMed comes in the following figure.



```
pubmed-DockingTit-set (0).txt - Notepad
File Edit Format View Help
PMID- 38495778
OWN - NLM
STAT- MEDLINE
DCOM- 20240319
LR - 20240319
IS - 2046-1402 (Electronic)
IS - 2046-1402 (Linking)
VI - 11
DP - 2022
TI - Bayesian Multi-View Clustering given complex inter-view structure.
PG - 1460
LID - 10.12688/f1000research.126215.2 [doi]
LID - 1460
AB - Multi-view datasets are becoming increasingly prevalent. These datasets consist
of different modalities that provide complementary characterizations of the same
underlying system. They can include heterogeneous types of information with
complex relationships, varying degrees of missingness, and assorted sample sizes,
as is often the case in multi-omic biological studies. Clustering multi-view data
allows us to leverage different modalities to infer underlying systematic
structure, but most existing approaches are limited to contexts in which entities
are the same across views or have clear one-to-one relationships across data
types with a common sample size. Many methods also make strong assumptions about
the similarities of clusterings across views. We propose a Bayesian multi-view
clustering approach (BMVC) which can handle the realities of multi-view datasets
that often have complex relationships and diverse structure. BMVC incorporates
known and complex many-to-many relationships between entities via a probabilistic
graphical model that enables the joint inference of clusterings specific to each
```

Figure 4.1 Raw Data

4.2 Extracting Useful Information

By applying the python script, the useful information was extracted which included PMID, Title, Abstract, Date of Publication, Place of Publication, and Journal of publication. Abstract and Title were used for finding the research trends present in the data. Others fields were extracted to find out other information that what year produced how many research papers or which country was at the top publishing the papers, in what journal most

papers were published, and the type of the article or research paper. By following figure, you can get the idea.

TITLE	ABSTRACT	DATE OF PUBLICATION	PUBLICATION TYPE	PLACE OF PUBLICATION	JOURNAL TITLE
Interleukin-6-Positive Immune Cells as a Possible New Immunologic Marker	Chronic inflammation creates tumor microenvironment (TME) that	2024 May-Jun 01	Journal Article	United States	Applied immunohistochemistry & molecular morphology : AIMM
Analyzing molecular signatures in preeclampsia and fetal growth	BACKGROUND: Intrauterine growth restriction (IUGR) and	2024	Journal Article	Switzerland	Frontiers in molecular biosciences
Effects of conceptus proteins on endometrium and blood leukocytes of	This study investigates the short and long-term effects of IFNT and	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
Single cell sequencing of zebrafish kidney marrows reveals AHR2-dependent	The aryl hydrocarbon receptor (AHR) is a ligand-dependent	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
Multiple direct and indirect roles of Paf1C in elongation, splicing, and	Paf1C is a highly conserved protein complex with critical	2024 Apr 25	Preprint	United States	bioRxiv : the preprint server for biology
Antibody-free time-resolved terbium luminescence assays designed for	Novel time-resolved terbium luminescence assays were developed	2024 Apr 24	Preprint	United States	bioRxiv : the preprint server for biology
Connecting Transcriptomics with Computational Modeling to Reveal	BACKGROUND: Nearly 1% or 1.3 million babies are born with	2024 Apr 23	Preprint	United States	bioRxiv : the preprint server for biology
Specific oncogene activation of the cell of origin in mucosal melanoma.	Mucosal melanoma (MM) is a deadly cancer derived from	2024 Apr 26	Preprint	United States	bioRxiv : the preprint server for biology
Single-nucleus and spatial landscape of the sub-ventricular zone in human	The sub-ventricular zone (SVZ) is the most well-characterized	2024 Apr 27	Preprint	United States	bioRxiv : the preprint server for biology
Cellular flows initiate left-right patterning prior to laterality gene	A bilateral body plan is predominant throughout the animal	2024 Apr 25	Preprint	United States	bioRxiv : the preprint server for biology
Development of Allosteric Small Molecule APOBEC3B Inhibitors from In	APOBEC3B cytosine deaminase contributes to the mutational	2024 Apr 25	Preprint	United States	bioRxiv : the preprint server for biology
Beyond A and B Compartments: how major nuclear locales define nuclear	Models of nuclear genome organization often propose a binary	2024 Apr 23	Preprint	United States	bioRxiv : the preprint server for biology
An integrated single-nucleus and spatial transcriptomics atlas reveals the	The hippocampus contains many unique cell types, which serve	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
Therapeutic Activity of Resolvin D1 (RvD1) in Murine MASH.	BACKGROUND AND AIMS: Recent studies have highlighted the	2024 Apr 26	Preprint	United States	bioRxiv : the preprint server for biology
The Role of X Chromosome in Alzheimer's Disease Genetics.	IMPORTANCE: The X chromosome has remained enigmatic in	2024 Apr 23	Preprint	United States	medRxiv : the preprint server for health sciences
Stochastic modelling of single-cell gene expression adaptation reveals	Cancer progression is an evolutionary process driven by the	2024 Apr 20	Preprint	United States	bioRxiv : the preprint server for biology
Sex-Specific Deflection of Age-Related DNA Methylation and Gene	BACKGROUND: Global and site-specific changes in DNA	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
Engineering a cell-free biosensor signal amplification circuit with	Cell-free systems are powerful synthetic biology technologies	2024 Apr 25	Preprint	United States	bioRxiv : the preprint server for biology
Transcriptomic characterization of human lateral septum neurons reveals	The lateral septum (LS) is a midline, subcortical structure, which	2024 Apr 26	Preprint	United States	bioRxiv : the preprint server for biology
Assessing drug uptake and response differences in 2D and 3D cellular	The architecture of cell culture-two-dimensional (2D) versus three-	2024 Apr 26	Preprint	United States	bioRxiv : the preprint server for biology
Genetic tuning of intrinsically photosensitive retinal ganglion cell subtype	The melanopsin-expressing, intrinsically photosensitive retinal	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
LHPP expression in triple-negative breast cancer promotes tumor growth	Triple-negative breast cancer (TNBC) is a highly aggressive and	2024 Apr 23	Preprint	United States	bioRxiv : the preprint server for biology
Droplet Hi-C for Fast and Scalable Profiling of Chromatin Architecture in	Comprehensive analysis of chromatin architecture is crucial for	2024 Apr 22	Preprint	United States	bioRxiv : the preprint server for biology
Predicting gene expression responses to environment in Arabidopsis	The evolution of gene expression responses are a critical	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
Genotype Representation Graphs: Enabling Efficient Analysis of Biobank-	Computational analysis of a large number of genomes requires a	2024 Apr 28	Preprint	United States	bioRxiv : the preprint server for biology
An integrated single-cell RNA-seq map of human neuroblastoma tumors and	Neuroblastoma is a common pediatric cancer, where preclinical	2024 Apr 22	Preprint	United States	bioRxiv : the preprint server for biology
Histopathological evaluation of abdominal aortic aneurysms with deep	Computational analysis of histopathological specimens holds	2024 Apr 24	Preprint	United States	medRxiv : the preprint server for health sciences
Design, synthesis, molecular docking, and molecular dynamic studies of	Introduction: Phosphodiesterase 7 (PDE7) is a high-affinity cyclic	2024	Journal Article	Switzerland	Frontiers in pharmacology
Enzymatic mechanism of MlrB for catalyzing linearized microcystins by	Microcystins (MCs) are the most widespread cyanobacterial	2024	Journal Article	Switzerland	Frontiers in microbiology
Amplifying endogenous stem cell migration for in situ bone tissue formation has	Endogenous stem cell-driven in situ bone tissue formation has	2024 Jun	Journal Article	England	Materials today. Bio
Genomic insights into the diversity, virulence, and antimicrobial resistance	INTRODUCTION: Detailed assessment of the population structure	2024	Journal Article	Switzerland	Frontiers in cellular and infection microbiology
Single-cell transcriptomics and Mendelian randomization reveal LUCAT1's	Background: Colorectal cancer (CRC) is a malignancy with high	2024	Journal Article	Switzerland	Frontiers in genetics

Figure 4.2 Dataset before cleaning

4.3 Cleaning Dataset

In figure 2 it can be seen that publication date is not equally distributed as some rows have only year and some have year and month while others have date, month and year which makes it difficult to get the results for a specific year so it was converted to only year. Also in the above dataset there was an issue with abstract and title which gave “\n” whenever there was a new line so this “\n” and any free spaces were removed. As it can be seen in the following figure.

PMID	TITLE	ABSTRACT	DATE OF PUBLICATION	PUBLICATION TYPE	PLACE OF PUBLICATION	JOURNAL TITLE
38144663	Alteration in serum miR126 expression in health	BACKGROUND: Fasting is practiced by various religions in the world	2023	Journal Article	India	Medical Journal, Armed Forces India
37645025	Differentiation and molecular characterization of	INTRODUCTION: Pluripotent stem cells have been used by various	2023	Journal Article	Iran	BioImpacts : BI
37090457	P(2)X and P2Y receptor antagonists reduce infla	BACKGROUND: P2 receptors have been implicated in the release of	2023	Journal Article	Spain	Pharmacy practice
36824519	Genomic identification, annotation, and compa	The hemipteran insect Diaphorina citri, or Asian citrus psyllid, is a v	2022	Journal Article	China	GigaByte (Hong Kong, China)
36777039	Singling out how genes are regulated during de	In this issue of Cell Genomics, McGarvey et al. characterize chroma	2022	News	United States	Cell genomics
36721815	Drug Repurposing for Identification of S1P1 Ago	Purpose: Drug repurposing is an approach successfully used for disc	2023	Journal Article	Iran	Advanced pharmaceutical bulletin
36721811	Effect of Nicotine on Immune System Function. A	As a parasympathetic alkaloid and the main substance in cigarette s	2023	Journal Article	Iran	Advanced pharmaceutical bulletin
36719113	Traditional herbal compounds as candidates to	COVID-19, a disease caused by the SARS-CoV-2 virus, is responsible	2023	Journal Article	England	Journal of biomolecular structure & dynamics
36699391	PERCC1, a new member of the Yap/TAZ/FAM181	MOTIVATION: Disrupted PERCC1 gene expression causes an interact	2022	Journal Article	England	Bioinformatics advances
36697791	How neocarcinand Octacid4 self-assembles with	Cram's supramolecular capsule Octacid4 can irreversibly and nonco	2022	Journal Article	England	Communications chemistry
36685757	Regulation of neuronal plasticity by the DNA re	Neurons respond rapidly to extracellular stimuli by activating signa	2022	Journal Article	Netherlands	Current research in neurobiology
36617427	CD171 Multi-epitope peptide design based on ti	Glioblastoma (GB) is a common primary malignancy of the central n	2023	Journal Article	England	Journal of biomolecular structure & dynamics
36617426	Computational designing of a novel subunit vac	Human cytomegalovirus (HCMV) is a widespread virus that can caus	2023	Journal Article	England	Journal of biomolecular structure & dynamics
36594555	Regulation of macrophage IFN β -stimulated gen	Macrophages serve as a first line of defense against microbial path	2023	Journal Article	England	Journal of cell science
36546249	Development of functional connectome gradien	Connectome mapping studies have documented a principal primar	2022	Journal Article	Netherlands	Science bulletin
36546013	A novel copper-sensing two-component system	In nature, bacteria must sense copper and tightly regulate gene exp	2022	Journal Article	Netherlands	Science bulletin
36513419	Comparison of dynamic 3D chromatin architect	INTRODUCTION: Heterosis is the major event driving plant develop	2022	Journal Article	Egypt	Journal of advanced research
36513418	A putative SUBTILISIN-LIKE SERINE PROTEASE 1	(INTRODUCTION: Panicle abortion is a severe physiological defect a	2022	Journal Article	Egypt	Journal of advanced research
36513416	GWAS identifies genetic loci underlying nitroge	INTRODUCTION: N responsiveness is the capacity to perceive and ir	2022	Journal Article	Egypt	Journal of advanced research
36513415	Genomic insights into the genetic signatures of	INTRODUCTION: Cultivated peanut (Arachis hypogaea L.) is an impo	2022	Journal Article	Egypt	Journal of advanced research
36474475	Osteoarthritis Data Integration Portal (OsteoDI	OBJECTIVE: OsteoDIP aims to collect and provide, in a simple search	2022	Journal Article	England	Osteoarthritis and cartilage open
36474464	Sustained delivery of the bone morphogenetic	OBJECTIVE: Attempts to utilise growth factors (GF) such as bone mo	2022	Journal Article	England	Osteoarthritis and cartilage open
36463507	LCMS-DNP based dereplication of Araucaria cun	As a part of natural defense, plants initiate the secretion of gum co	2022	Journal Article	England	Natural product research
36419596	Alteration of hemoglobin β gene expression in	Hemoglobin beta (Hb β) is a heme-binding protein capable of oxyg	2022	Journal Article	England	Fish and shellfish immunology reports
36411529	Study of potent CDK7 inhibitor secondary metal	Chemical investigation of the petroleum ether extract of heartwoo	2022	Journal Article	England	Natural product research
36406284	[Cu(dipicolinoylamide)(NO $_3$)](H $_2$ O)] as anti-4	For first time the new N-picolinoypicolinamide was obtained as in	2022	Journal Article	Netherlands	Journal of molecular structure
36386473	Biosynthesis of rumbirins and inspiration for di	Investigation on how nature produces natural compounds with che	2022	Journal Article	Netherlands	Acta pharmaceutica Sinica. B
36366999	One-year surveillance of SARS-CoV-2 in wastew	The current COVID-19 pandemic has emphasized the vulnerability of	2022	Journal Article	England	Journal of water and health
36366997	Molecular characterization and phylogenetic an	Free-living amoebae (FLA) are protozoa ubiquitously found in natu	2022	Journal Article	England	Journal of water and health

Figure 4.3 Dataset

4.4 Applying Algorithms after Preprocessing

After the data pretreatment stages were finished, the stage was set for the use of topic modeling approaches to discover the latent thematic patterns hidden in the study article corpus. Topic modeling, an unsupervised machine learning technique, seeks to automatically identify abstract "topics" that run through a collection of documents using statistical patterns of keyword co-occurrence.

This section delves into the implementation and evaluation of two fundamental topic modeling algorithms: LDA and NMF. By applying these methods to the preprocessed corpus, valuable insights were gained regarding the prevalent concepts, their contextual relationships, and the underlying semantic structures inherent to the domain of research encompassed by the articles.

4.5 Exploratory Data Analysis (EDA)

4.5.1 Distribution of Publications Across Years

By the EDA it is clear that for the terms which were searched on PubMed are in trend. The quantity of research papers increases with the passage of time.

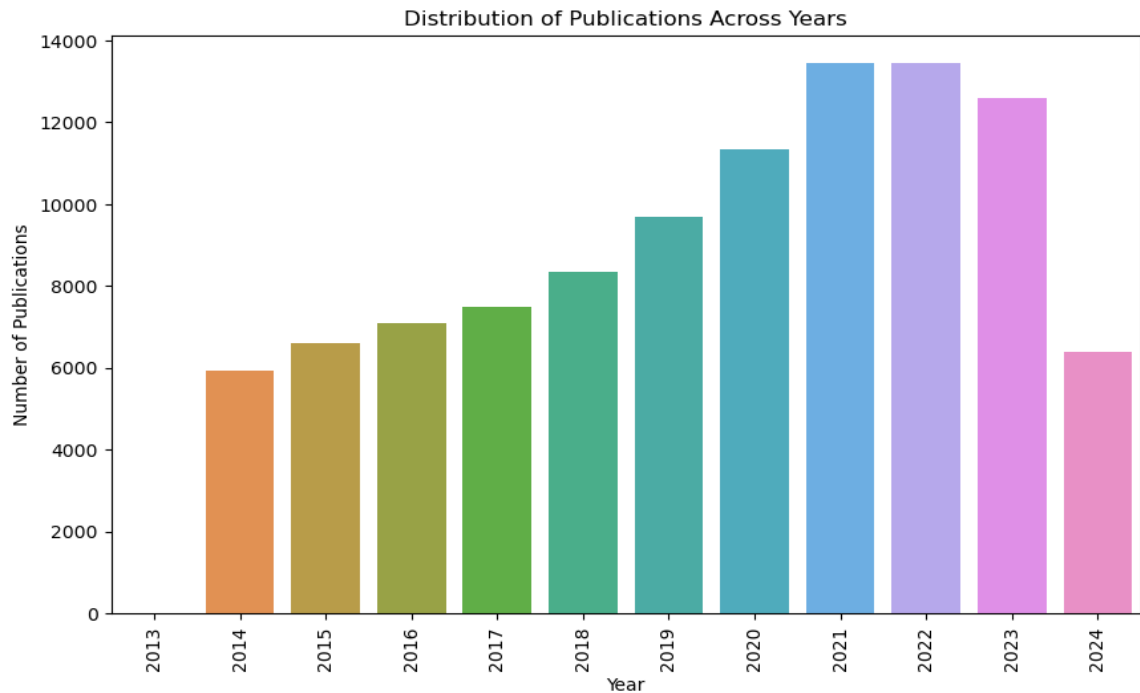


Figure 4.4 Distribution of Publications Across Years

4.5.2 Top Countries of Publication

The most number of research articles published in England and then on second number was United State and then Switzerland and Netherlands. Most of the research papers were submitted in England and United States.

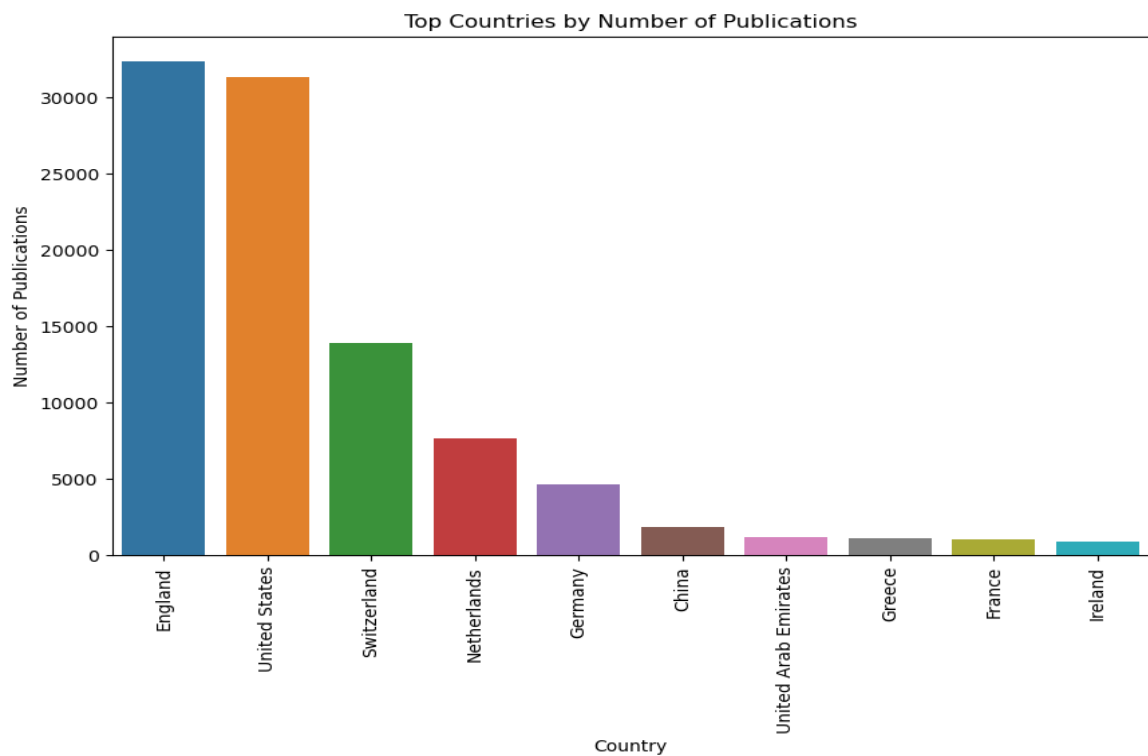


Figure 4.5 Top Countries of Publication

4.5.3 Top Journals

Following is contribution of the top journals around the word in publishing the articles in the PubMed database.

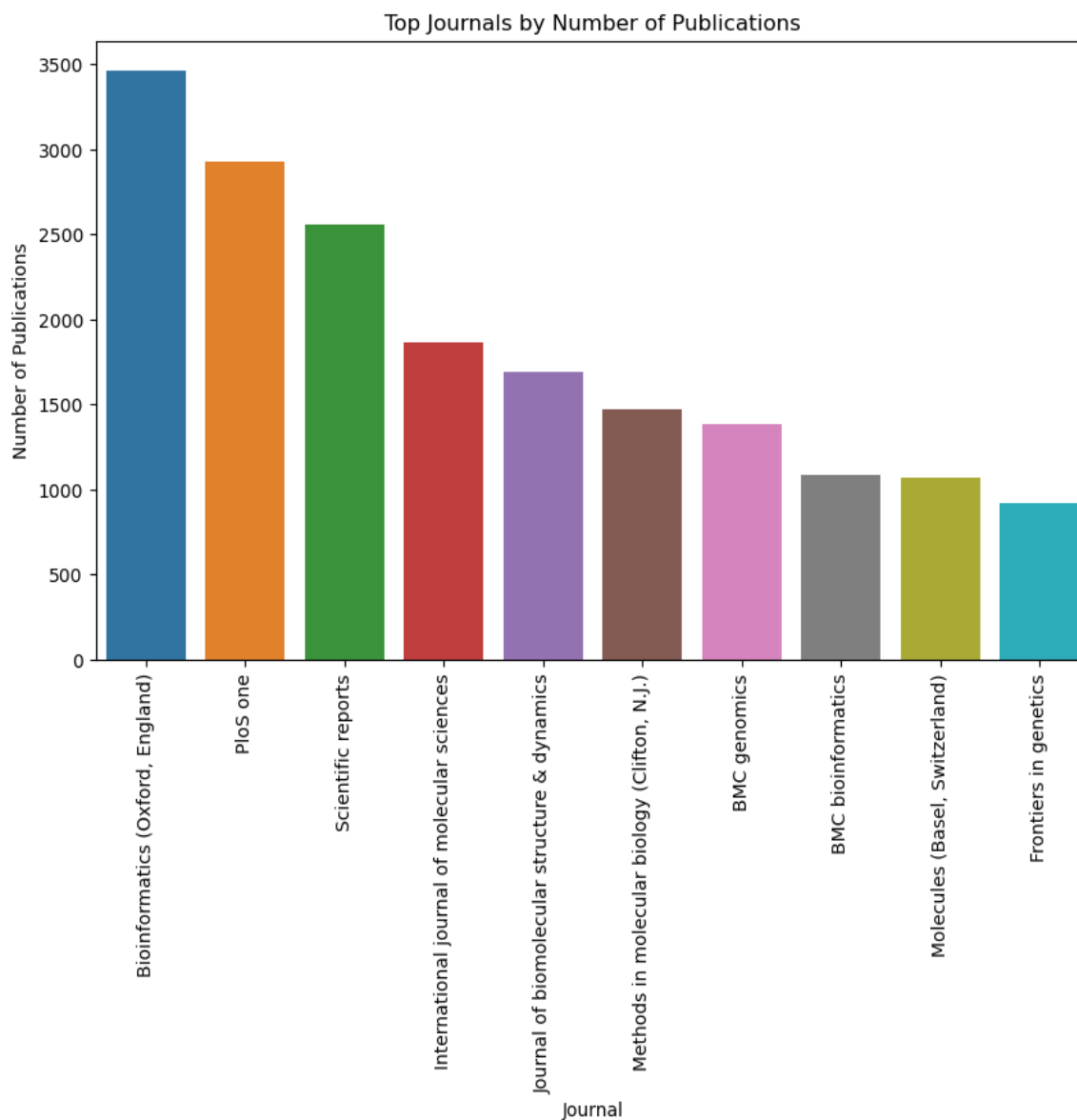


Figure 4.6 Top Journals

4.5.4 Top Article Types

It was also found that most abundant articles were journal articles which were around 94% of the whole articles. Then there are other types of articles like case reports, comparative study evaluation study, English abstract, preprint, clinical trial, editorial, letter, and comment.

Top Article Types

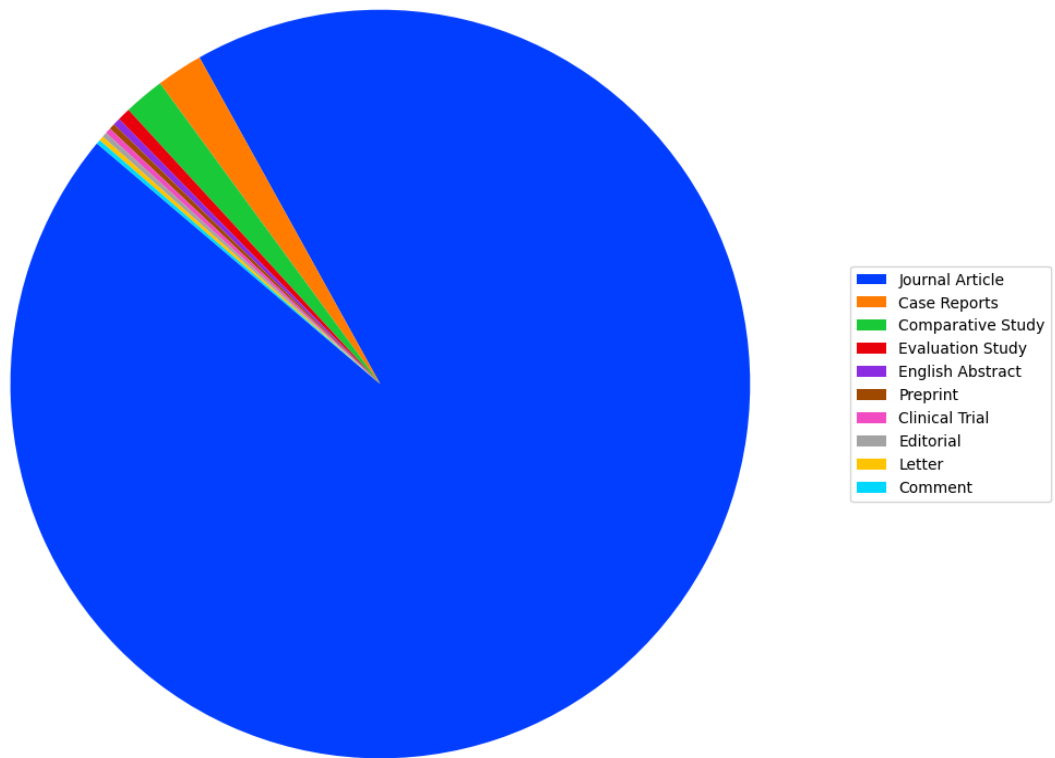


Figure 4.7 Top Article Types

4.5.5 Most Frequent Words before Topic Modeling

Before immersing into the topic modeling results, it is instructive to examine the 20 most frequently occurring words within the preprocessed research article corpus. This vocabulary prevalence overview provides a foundational understanding of the dominant lexical elements and offers context for interpreting the subsequent topic modeling outcomes.

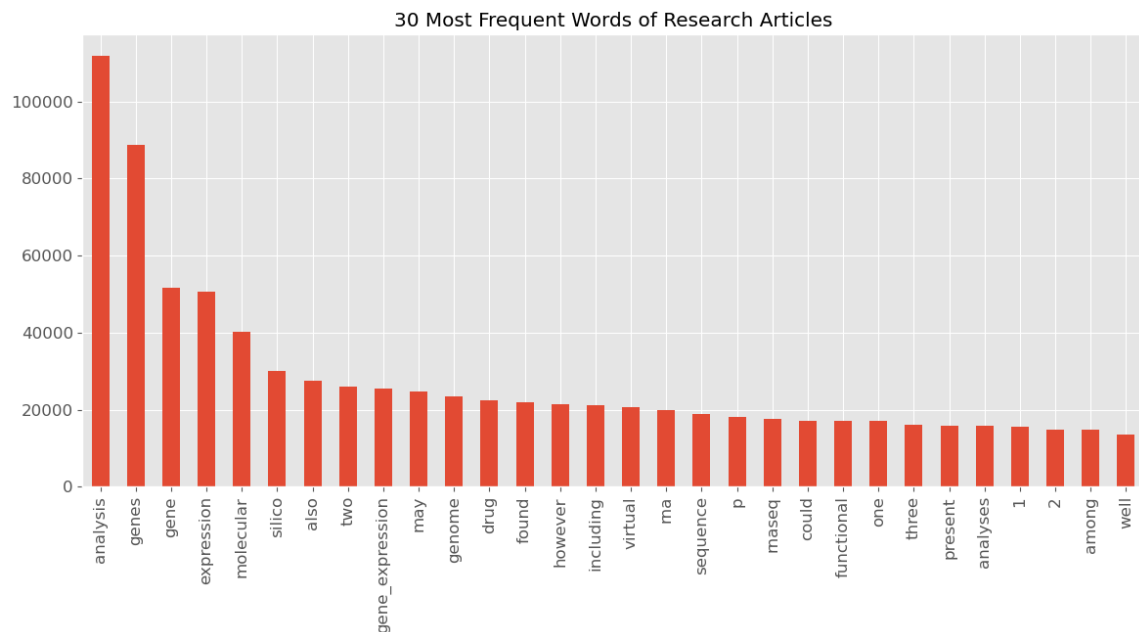


Figure 4.8 Most Frequent Words

The image above depicts a bar chart representing the 20 most frequently used words across the entire corpus, with the x-axis indicating the word frequencies. As expected, the most prevalent terms encompass commonly employed words in scientific writing, such as "analysis", "genes", "expression", "molecular" and "silico". While the dominance of these high-frequency words is unsurprising given the academic nature of the corpus, their ubiquity highlights the inherent challenge in discerning the underlying conceptual structures solely based on surface-level lexical patterns.

It is precisely this limitation that underscores the necessity and power of topic modeling algorithms. By leveraging advanced statistical techniques, these algorithms can transcend the explicit word co-occurrence patterns and uncover the latent thematic compositions that characterize the corpus. The subsequent sections detail the application and evaluation of two prominent topic modeling approaches, LDA and NMF, on the research article data.

4.6 Latent Dirichlet Allocation (LDA)

LDA is a multiplicative probabilistic model that has emerged as a widely adopted and influential technique for topic modeling. Its fundamental assumption is that each document consists of numerous topics in different quantities, while every topic itself is a sharing over the entire vocabulary. In essence, LDA posits that documents are comprised of a mixture of latent topics, and each word within a document is attributable to one of these topics (Bastani *et al.*, 2019).

The objective of LDA is to automatically discover these latent topics and their corresponding word distributions based on the statistical co-occurrence patterns within the corpus. By analyzing the word frequencies and co-occurrences across documents, LDA can infer the most probable topic distributions for each document and the most salient words associated with each topic.

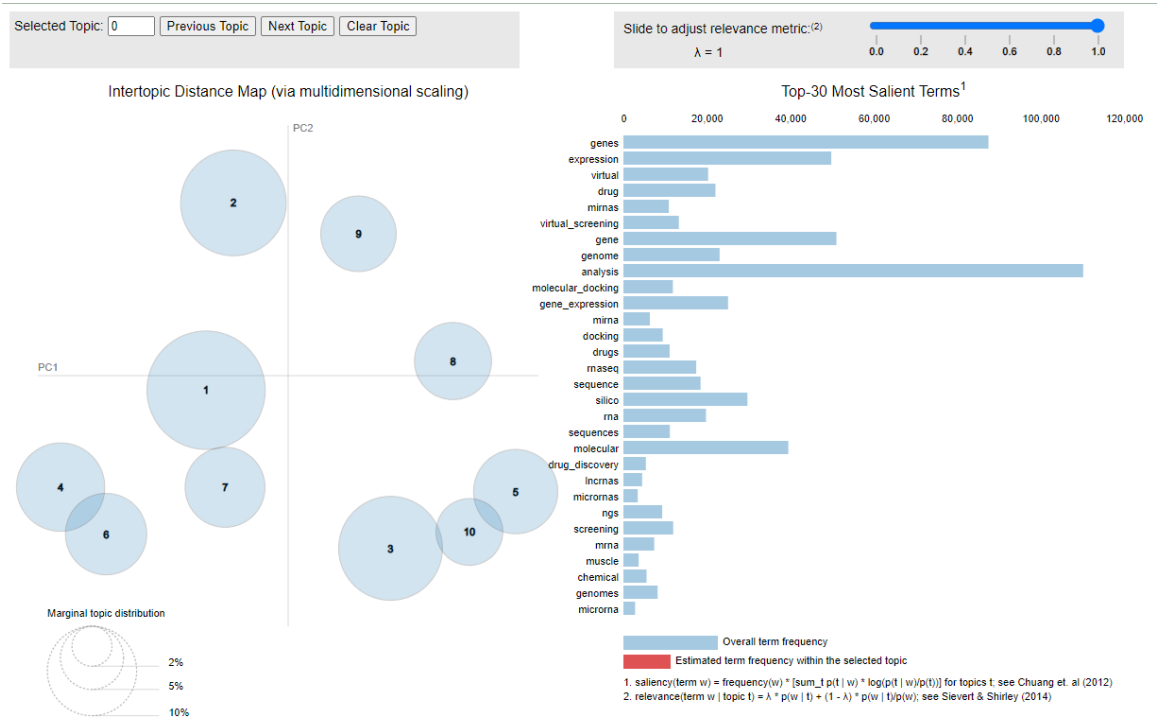


Figure 4.9 LDA Implementation

The interactive LDA visualization above provides a comprehensive, visually engaging representation of the most prominent topics discovered by the algorithm within the research article corpus. Each bubble represents a distinct topic, with its area proportional to the overall prevalence or importance of that topic across the entire corpus. The visualization enables seamless exploration of the associations between topics and their top representative words, as well as the distribution of topics across individual documents. To quantitatively evaluate the performance and quality of the LDA model, two metrics were computed: coherence score and perplexity. The coherence score assesses the semantic coherence or interpretability of the identified topics, with higher values indicating more cohesive and meaningful topics. Specifically, it measures the degree to which the top words within a topic co-occur frequently within the same document, thereby capturing the semantic relatedness and consistency of the topic's constituent words.

```
# Calculate Coherence Score for LDA
texts = [text.split() for text in data['text']]
id2word = corpora.Dictionary(texts)
corpus = [id2word.doc2bow(text) for text in texts]
lda_model = gensim.models.ldamodel.LdaModel(corpus=corpus, id2word=id2word, num_topics=10, random_state=42)
coherence_model_lda = CoherenceModel(model=lda_model, texts=texts, dictionary=id2word, coherence='c_v')
coherence_lda = coherence_model_lda.get_coherence()
print(f'Coherence Score for LDA: {coherence_lda}')
```

Coherence Score for LDA: 0.47347558747720314

Figure 4.10 LDA Coherence Score

The coherence score obtained for the LDA model on this research article corpus is 47%. This suggests that the topics identified by LDA exhibit moderate semantic coherence, implying that the word distributions within each topic are moderately related and interpretable. The perplexity metric, on the other hand, measures the inverse likelihood of the LDA model generating the observed data. In simpler terms, it quantifies how well the model can predict or generalize to unseen documents, with lower perplexity values indicating better generalization performance.

The perplexity value obtained for the LDA model on this corpus is -9.472. This suggests that the model's ability to generalize and predict unseen documents is high, indicating good overall model fit and generalization performance.

While these quantitative metrics provide valuable insights into the LDA model's quality and suitability for the research article corpus, a more intuitive understanding of the derived topics can be gained through visual representations and word listings.

Table 4.1 LDA Topics

Topic No.	LDA Topics
<i>Topic 1</i>	virtual screening ngs medicine practice nextgeneration primary analysis including test
<i>Topic 2</i>	genes analysis expression gene gene_expression molecular analyses including normal analyzed
<i>Topic 3</i>	mirnas expression mirna analysis genes micrnas gene_expression muscle micrna injury
<i>Topic 4</i>	drug molecular silico virtual_screening molecular_docking drugs docking analysis screening drug_discovery
<i>Topic 5</i>	analysis functional gene diseases silico function heart cardiac neural disorders
<i>Topic 6</i>	expression genes analysis gene rnaseq rna gene_expression Incrnas differentiation mrna
<i>Topic 7</i>	virtual gene analysis type coronary values case congenital nov autosomal
<i>Topic 8</i>	genes analysis gene expression rnaseq gene_expression rice factors conserved tolerance
<i>Topic 9</i>	analysis genome sequence sequences gene genes rna nextgeneration genomes ngs
<i>Topic 10</i>	analysis silico molecular structural membrane effect mass vivo antiinflammatory antioxidant

Each topic is represented by a list of its top 10 most relevant or salient words, ranked in descending order of importance or association strength. These word lists offer a concise yet comprehensive view of the thematic compositions identified by LDA, enabling a more nuanced interpretation of the underlying concepts and their contextual relationships.

For instance, Topic 1 appears to be centered around Virtual Screening and Next-Generation Sequencing, while Topic 5 seems to capture the theme of Functional Analysis in Cardiac Disorders. By carefully examining these word lists in conjunction with the word cloud and

interactive visualization, a rich tapestry of thematic structures emerges, providing valuable insights into the conceptual landscape of the research article corpus.

4.7 Non-negative Matrix Factorization (NMF)

Complementing the LDA analysis, NMF was employed as an alternative topic modeling technique. Unlike LDA's generative probabilistic method, NMF is a dimensionality reduction and factorization of matrix algorithm that breaks down the document-term matrix into two low-rank matrix structures: one represents the topics and the other captures the document-topic distributions. (Gan *et al.*, 2021).

The key advantage of NMF lies in its ability to extract interpretable and additive topic representations, as well as its inherent sparsity constraints that promote more concentrated and discriminative topics. Additionally, NMF's matrix factorization formulation enables the identification of document clustering patterns based on their topical compositions, offering a unique perspective on the data.

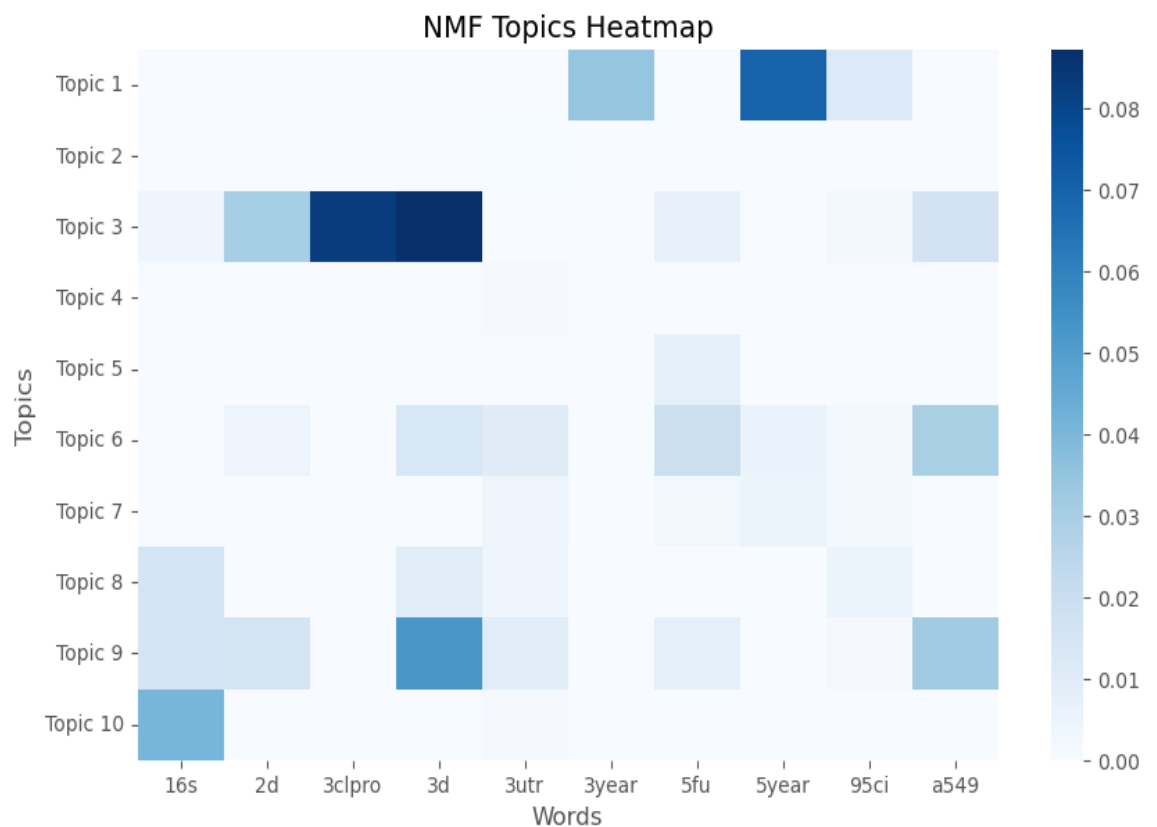


Figure 4.12 NMF Heatmap

The heatmap above visualizes the document-topic distributions obtained from the NMF analysis on the research article corpus. Each row represents an individual document, each

column represents a distinct topic, and the color intensity reflects the degree of association or membership between the document and the corresponding topic. Darker shades indicate stronger associations, while lighter shades represent weaker or negligible connections.

This visualization facilitates the identification of topic prevalence patterns across the corpus, as well as the exploration of document clustering based on their shared topical compositions. Documents exhibiting similar topic distributions tend to be grouped together, enabling the discovery of thematic similarities and potential sub-domains within the broader research area encompassed by the corpus.

To further elucidate the thematic structures uncovered by NMF, a series of topic visualization images are provided below, depicting the top words associated with each topic and their relative importance or salience.

Table 4.2 NMF Topics

Topic No.	NMF Topics
<i>Topic 1</i>	genes analysis gene_expression gene kyoto microarray encyclopedia coexpression cytoscape analyses
<i>Topic 2</i>	virtual_screening molecular molecular_docking docking silico ligand site structurebased kcalmol hits
<i>Topic 3</i>	genome sequence ngs sequences nextgeneration genomes analysis genomics generation reference
<i>Topic 4</i>	mirnas mirna micrnas micrna expression mrnas rnas mrna circrnas mirnamrna
<i>Topic 5</i>	virtual screening test interventions anxiety practice physical stroke objective healthcare
<i>Topic 6</i>	rna rnaseq circrnas rnas scrnaseq circular gene_expression analysis circrna protocol
<i>Topic 7</i>	drug drugs drug_discovery silico discovery screening diseases coronavirus recent effective
<i>Topic 8</i>	expression analysis adenocarcinoma biomarker microenvironment gene_expression gc immunotherapy os poor
<i>Topic 9</i>	lncrnas lncrna rnas mrnas cerna rna coexpression expression roles networks
<i>Topic 10</i>	gene analysis silico expression functional drought function rice promoter type

These topic images offer an intuitive and visually engaging representation of the topics derived by NMF. The size and positioning of the words within each image convey their relative importance and semantic associations within the respective topic. Larger words located centrally within the image are considered more salient or representative of the underlying theme, while smaller words on the periphery play a lesser role in defining the topic's conceptual composition.

```
# Calculate Coherence Score for NMF
nmf_model_gensim = gensim.models.nmf.Nmf(corpus=corpus, num_topics=10, id2word=id2word, random_state=42)
coherence_model_nmf = CoherenceModel(model=nmf_model_gensim, texts=texts, dictionary=id2word, coherence='c_v')
coherence_nmf = coherence_model_nmf.get_coherence()
print(f'Coherence Score for NMF: {coherence_nmf}')
```

Coherence Score for NMF: 0.4838742764620447

Figure 4.13 NMF Coherence Score

The coherence score obtained for the NMF model on this research article corpus is 48%. This suggests that the topics identified by NMF exhibit strong semantic coherence, implying that the word distributions within each topic are moderately related and interpretable.

4.8 LDA and NMF Comparison

For example, the topic image for Topic 1 in LDA is dominated by large, centrally positioned words such as virtual, screening, ngs, medicine, practice, and nextgeneration suggesting that this topic is primarily focused on the theme of Virtual Screening and Next-Generation Sequencing. For NMF, topic 1 is dominated by genes, analysis, gene_expression, gene, Kyoto, microarray, and encyclopedia. Conversely, the topic image for Topic 2 exhibits a more dispersed distribution of words, for LDA with genes, analysis, expression, gene, gene_expression, molecular, analyses and for NMF with virtual_screening, molecular, molecular_docking, docking, silico, and ligand being the most prominent, potentially indicating a broader or more diverse thematic scope encompassing gene expression and molecular docking or virtual screening respectively. Following is the comparison given by the topics obtained from LDA and NMF (Gurcan, 2022).

Table 4.3 LDA and NMF Comparison

Topic No.	LDA	NMF
1	virtual screening ngs medicine practice nextgeneration	genes analysis gene_expression gene kyoto microarray encyclopedia
2	genes analysis expression gene gene_expression molecular	virtual_screening molecular molecular_docking docking silico ligand
3	mirnas expression mirna analysis genes micrnas gene_expression	genome sequence ngs sequences nextgeneration genomes analysis genomics gene
4	drug molecular silico virtual_screening molecular_docking drugs docking drug_discovery	mirnas mirna micrnas micrna expression mrnas rnas mrna circrnas mirnamr
5	analysis functional gene diseases silico function heart cardiac	virtual screening test interventions anxiety practice physical
6	expression genes analysis gene rnaseq rna gene_expression lncrnas	rna rnaseq circrnas rnas scrnaseq circular gene_expression analysis circrna
7	virtual gene analysis type coronary values case	drug drugs drug_discovery silico discovery screening diseases
8	genes analysis gene expression rnaseq gene_expression	expression analysis adenocarcinoma biomarker microenvironment gene_expression
9	analysis genome sequence sequences gene genes rna nextgeneration	lncrnas lncrna rnas mrnas cerna rna coexpression expression
10	analysis silico molecular structural membrane effect mass	gene analysis silico expression functional drought function

The word cloud generated from the topics extracted using LDA illustrated a focus on several key themes in the research corpus.

indicating a trend towards understanding genetic factors in health and disease. The repeated mention of "gene" and "analysis" reflects the growing importance of transcriptomics in elucidating gene function and regulation.

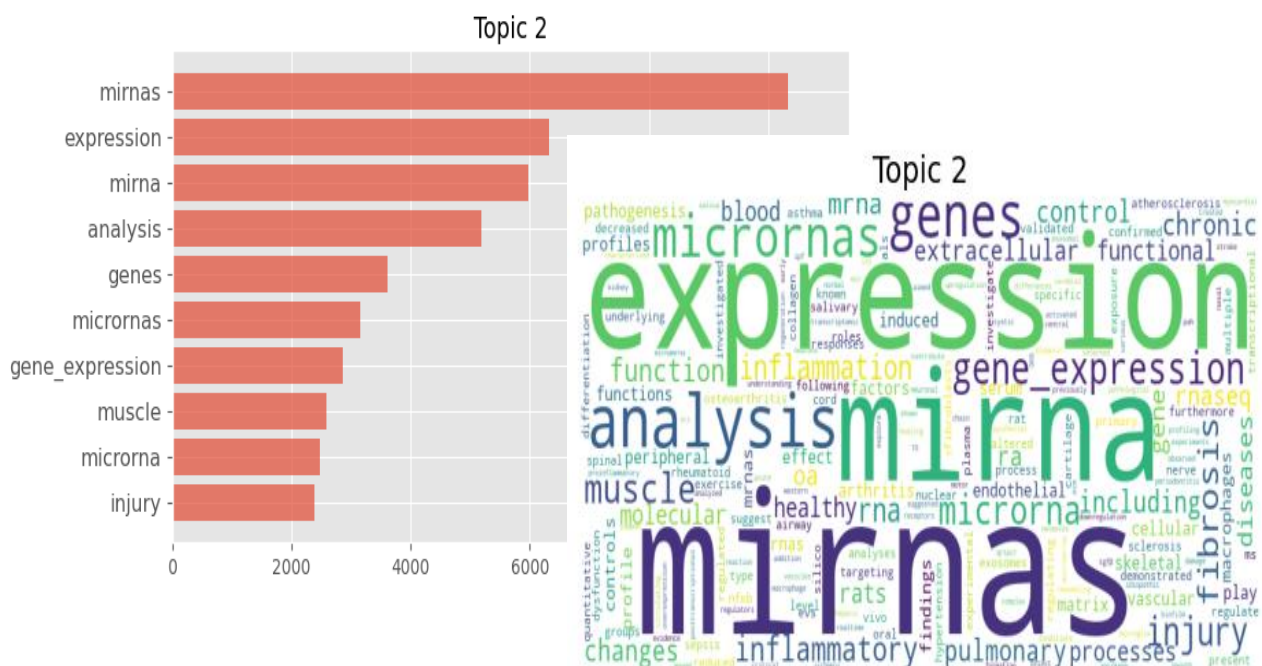


Figure 4.16 Topic 3 for LDA

The word cloud generated from the topics extracted using LDA highlighted a focus on miRNA expression and gene analysis. Key terms such as "miRNAs", "expression", and "microRNAs" were prominently featured, indicating a significant emphasis on studying microRNAs and their regulatory roles in gene expression. Terms like "analysis," "genes," and "muscle" suggested research involving the impact of miRNAs on muscle-related processes. The presence of words like "injury" and "gene_expression" pointed to investigations on miRNA's role in muscle injury and recovery.

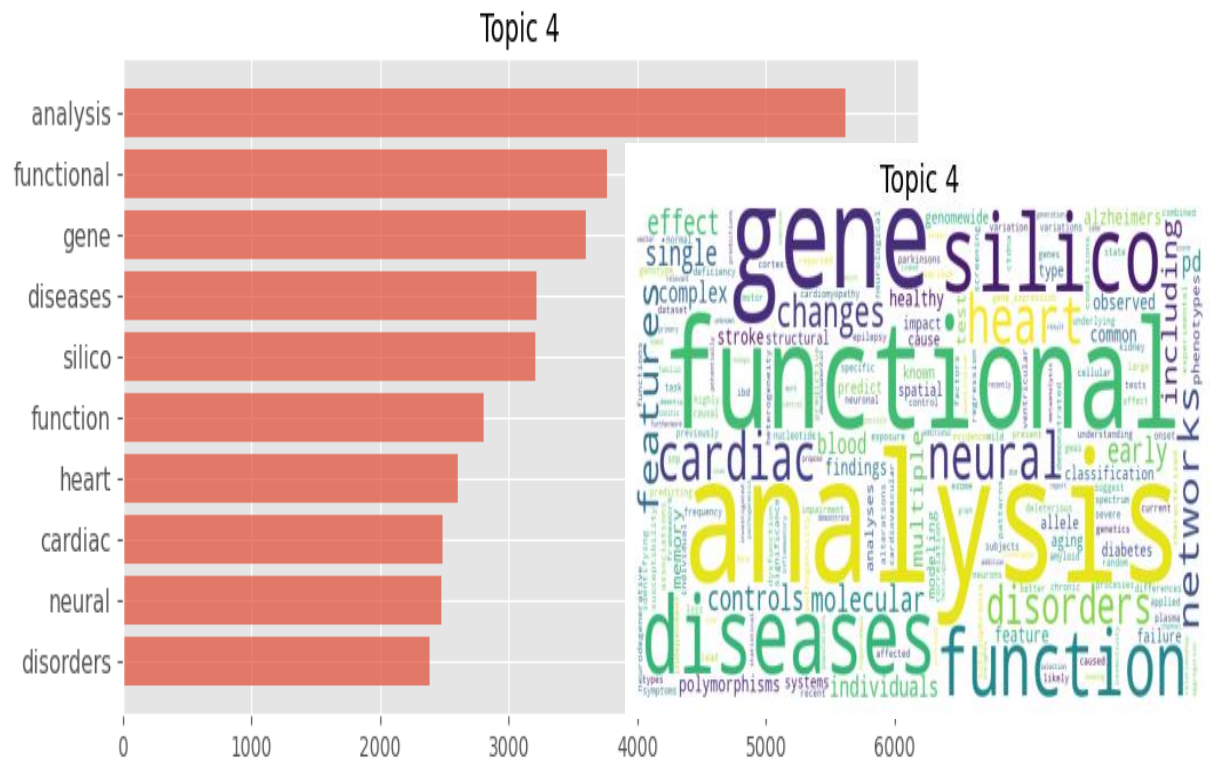


Figure 4.18 Topic 5 for LDA

The word cloud generated from the topics extracted using LDA emphasized research in gene function, diseases, and in silico methods. This topic addresses the functional analysis of genes related to cardiac disorders, suggesting a trend towards understanding the genetic basis of heart diseases. The inclusion of terms like "neural disorders" indicates a potential overlap with neurocardiology. Prominent terms such as "analysis", "functional", and "gene" suggested a focus on the role of genes in disease mechanisms. Key topics included "heart", "cardiac", and "neural," indicating studies on heart and neural disorders. The presence of "diseases" and "disorders" pointed to research on various medical conditions, while "function" and "silico" highlighted the use of computational methods to explore gene functions in these contexts.

The word cloud generated from the topics extracted using NMF illustrated a focus on several key themes in the research corpus.

resources such as KEGG and Cytoscape for deeper understanding of gene networks and functional genomics.

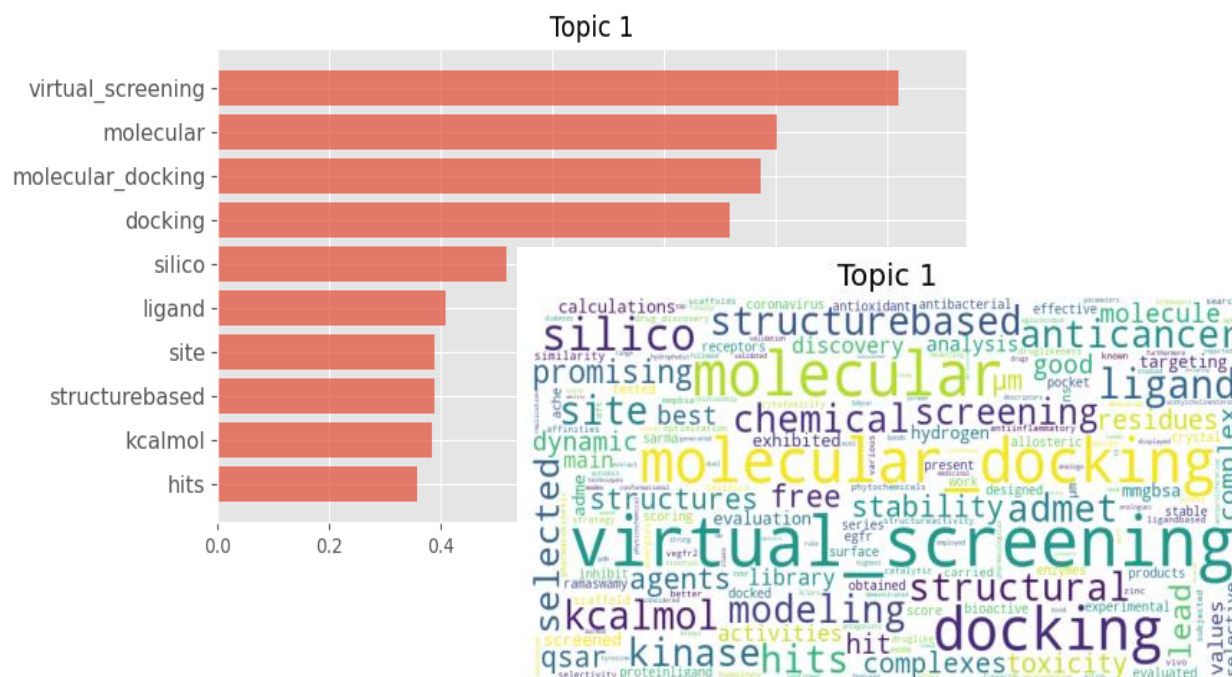


Figure 4.20 Topic 2 for NMF

The word cloud presented a visual representation of topics extracted using a topic modeling technique for Topic 2. The dominant themes that emerged were centered around virtual screening, molecular docking, and in silico methods. The most prominent words in the cloud were "virtual_screening", "molecular", "docking", and "ligand," indicating these were core concepts of the topic. The word cloud also highlighted related terms such as "site", "structurebased", and "kcalmol," suggesting a focus on computational techniques in drug discovery, particularly identifying potential ligand-receptor interactions and docking results.

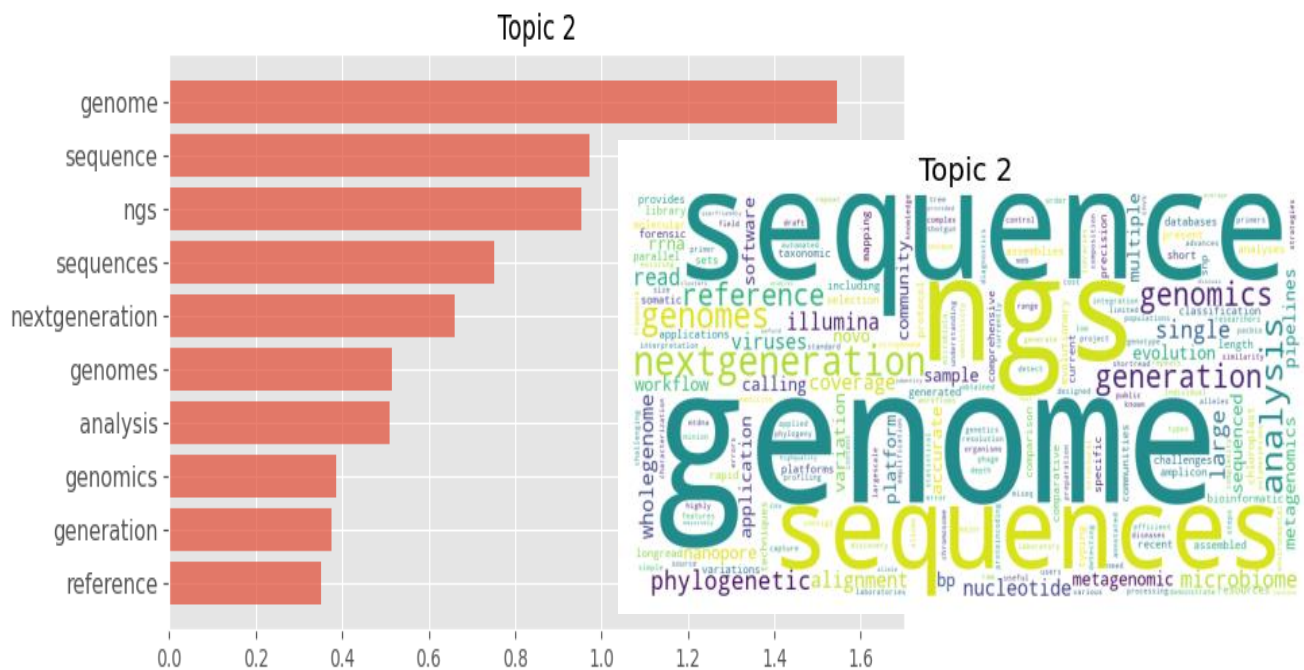


Figure 4.21 Topic 3 for NMF

The word cloud presented a visual representation of topics extracted for Topic 3, using a topic modeling technique such as NMF. The dominant themes that emerged from this visualization were centered around genome analysis and next-generation sequencing (NGS). The most prominent words in the cloud were "genome", "sequence", "NGS", and "reference", indicating these were the core concepts of the topic. The word cloud also highlighted related terms like "genomes", "sequences", and "genomics", suggesting a focus on NGS methods for studying genomes and genomic variations.

Overall, this topic appeared to encompass a wide range of approaches to genomics and NGS, including reference genome generation, sequence analyses, and applications of these methods in identifying genetic variations for various research and clinical purposes.

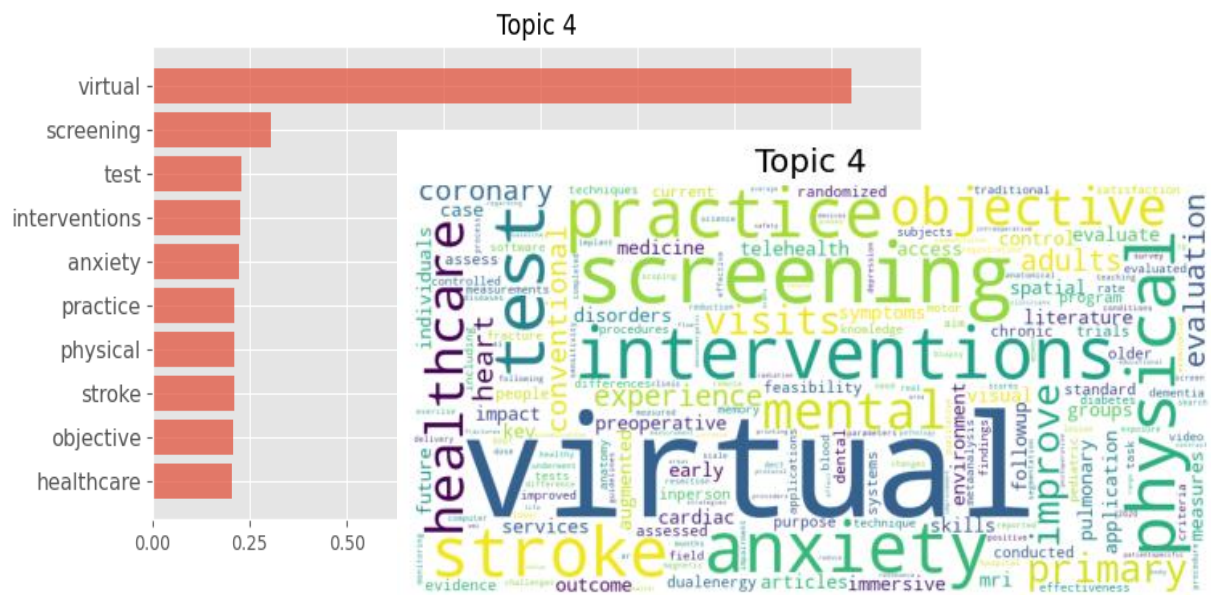


Figure 4.23 Topic 5 for NMF

The image showed a word cloud representing topics extracted using Non-negative Matrix Factorization (NMF). The most prominent terms in the word cloud were "virtual screening", "interventions", "anxiety", and "healthcare," suggesting these were central themes. The word cloud appeared to represent topics in healthcare research, with a particular emphasis on interventions related to anxiety, physical health, and stroke management. It suggested a comprehensive view of studies focused on diagnostic practices and therapeutic approaches within the context of healthcare objectives (Shah *et al.*, 2023).

CHAPTER NO 5

SUMMARY

This research investigates the application of topic modeling techniques to a corpus of research articles from the scientific domain of bioinformatics. The overarching goal is to uncover the latent thematic structures and conceptual compositions that underlie the body of literature encompassed by the corpus. By leveraging advanced machine learning algorithms and natural language processing techniques, valuable insights can be gained into the prevalent research topics, their contextual relationships, and the evolution of ideas within bioinformatics over time.

The research methodology employed a systematic approach, beginning with the curation and preprocessing of the research article corpus. Data preprocessing steps, such as tokenization, stopwords removal, and stemming/lemmatization, were applied to transform the raw textual data into a structured format suitable for topic modeling algorithms. Additionally, an analysis of the 20 most frequent words within the preprocessed corpus was conducted, providing a baseline understanding of the dominant vocabulary and highlighting the need for more sophisticated techniques to uncover the underlying conceptual themes.

Following the data preprocessing stage, two fundamental topic modeling algorithms were employed: Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). LDA, a generative probabilistic model, assumes that each document exhibits multiple topics in varying proportions, and each topic is a distribution over the entire vocabulary. By analyzing word co-occurrence patterns, LDA can infer the most probable topic distributions for each document and the most salient words associated with each topic. The application of LDA to the research article corpus yielded an interactive visualization depicting the most prominent topics, their top representative words, and their relative prevalence across the corpus. Quantitative metrics, including coherence score and perplexity, were computed to evaluate the LDA model's performance and quality. Additionally, a word cloud and tabular representation of the top 10 topics with their 10 most relevant words were provided, offering intuitive visualizations and granular insights into the thematic structures uncovered by LDA.

Complementing the LDA analysis, Non-negative Matrix Factorization (NMF) was employed as an alternative topic modeling technique. NMF is a dimensionality reduction and matrix factorization algorithm that decomposes the document-term matrix into two

low-rank matrices, representing the topics and document-topic distributions. A heatmap visualization was generated, depicting the document-topic associations and facilitating the identification of document clustering patterns based on shared topical compositions.

Furthermore, a series of topic visualization images were produced, illustrating the top words associated with each NMF-derived topic and their relative importance or salience. These visual representations offered an engaging and interpretable perspective on the thematic structures uncovered by NMF within the research article corpus.

In addition to the topic modeling analyses, further exploration was conducted to understand the distributions and trends present in the corpus across various dimensions, including publication year, geographic location, journal sources, and publication types. Visualizations such as bar charts, pie charts, and histograms were employed to illustrate these distributions, providing valuable contextual information and enabling the identification of potential patterns or biases within the data. The research findings have significant implications for the scientific community and academic researchers. By uncovering the latent thematic structures and conceptual compositions within the research literature, this study offers a comprehensive understanding of the intellectual landscape, facilitating the identification of emerging topics, research gaps, and potential areas for future exploration. Additionally, the insights gained from this research can inform the development of more effective information retrieval systems, literature recommendation engines, and knowledge management strategies within academic and research institutions. Furthermore, the methodology and techniques employed in this study can be adapted and applied to other domains or literature corpora, enabling the extraction of valuable knowledge and insights across various fields of study. By leveraging the power of topic modeling algorithms and data visualization techniques, researchers can gain a deeper understanding of the conceptual structures underlying their respective domains, fostering interdisciplinary collaborations and driving the advancement of knowledge. Following were the main topics discovered by LDA and NMF.

Gene expression: Gene expression analysis plays a central role in cancer research and personalized medicine. Both models highlight a strong emphasis on the genetic basis of cancer, focusing on gene expression patterns and mutations. This indicates a significant trend towards understanding how gene expression influences tumor behavior and patient outcomes. Key terms such as "gene expression," "cancer," "tumor," "mutations," and "prognostic" frequently appear across various topics, underscoring the importance of gene expression in developing tailored treatment strategies for cancer patients.

Drug Discovery and Molecular Docking: A major trend evident in the results is the use of computational methods for drug discovery and design. Both LDA and NMF models show topics related to molecular docking, virtual screening, and in silico drug design. This indicates a growing emphasis on using bioinformatics tools to accelerate the drug development process and identify potential therapeutic targets.

Next-Generation Sequencing and Genomic Analysis: The prominence of terms like "sequencing," "genome," "ngs" (next-generation sequencing), and "genomic" across multiple topics in both models points to the continued importance of high-throughput sequencing technologies in bioinformatics research. This trend encompasses various applications, from whole-genome sequencing to transcriptomics and epigenomics studies.

These themes reflect current research priorities in bioinformatics, focusing on translating genomic insights into clinical applications, leveraging computational power for drug discovery, and advancing our understanding of complex biological systems through large-scale sequencing and analysis techniques.

Overall, this research contributes to the field of text mining, natural language processing, and knowledge discovery by demonstrating the efficacy of topic modeling techniques in uncovering the thematic compositions and contextual relationships within large textual corpora. The findings and methodologies presented in this study have the potential to enhance academic research, knowledge management practices, and interdisciplinary collaborations, ultimately propelling the frontiers of scientific inquiry and intellectual discourse.

The LDA and NMF analyses reveal significant trends in bioinformatics research, highlighting the importance of gene expression, molecular docking, and NGS technologies. The coherence scores indicate a good quality of topics derived from the models, reflecting meaningful patterns within the dataset. Future research can build upon these findings by exploring the identified topics in greater depth and their implications in the field of bioinformatics.

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